# SMALL NON-STRONG I-CONTRACTIBLE GRAPHS AND ODD TORSION IN FLAG COMPLEXES

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ABSTRACT. We construct three 14-vertex I-contractible graphs that are not strong I-contractible, and three 13-vertex $\mathbb{Z}_2$-acyclic graphs whose clique complexes have $H_1 \cong \mathbb{Z}/3$. These examples refute Conjectures 14 and 15 of Dochtermann, Espinoza, Frías-Armenta and Hernández [Discrete Math. 346 (2023) 113516]. Their homology and contractibility certificates are checked independently. A reduction to graphs admitting no strong vertex or edge deletion makes an exhaustive search feasible. The recorded searches examined 21,011,415,992,470 graphs on 13 vertices and 64,856,470,381,820 $K_4$-free graphs on 14 vertices. Subject to the explicit computational hypotheses stated in the paper, the respective minimum orders are 13 and 14, and there are exactly eight $K_4$-free graphs of order 14 in the gap between I-contractibility and strong I-contractibility. The three 13-vertex examples attain the minimum order for odd torsion in the first homology of a flag complex, using Adamaszek's classification for at most 12 vertices. We also report cardinalities through order 13, explain the remaining certification gaps in the enumeration, and provide graph data, code and finite certificates.

## 1. INTRODUCTION

Ivashchenko's contractible transformations delete or add vertices and edges subject to a contractibility condition on their neighbourhoods [1, 2, 3]. A graph is *I-contractible* if it belongs to the class $\mathfrak{I}$ generated from a single vertex by these transformations. The relation with the topology of its clique complex $\Delta(G)$ is now precise: Chen, Yau and Yeh [4] established the connection with simple homotopy, and Dochtermann and Matsushita [5] proved that I-homotopy of graphs is equivalent to simple homotopy equivalence of clique complexes. In particular, $G \in \mathfrak{I}$ if and only if $\Delta(G)$ is contractible.

Restricting the available transformations gives the classes $\mathfrak{I}_{VS}$ and $\mathfrak{I}_S$ of vertex-strong and strong I-contractible graphs [6]. Their definitions require a successful sequence of vertex deletions, or of vertex and edge deletions, respectively; the relevant neighbourhoods must themselves belong to the same recursive class. Thus

$$\mathfrak{I}_{VS} \subseteq \mathfrak{I}_S \subseteq \mathfrak{I} = \{G : \Delta(G) \text{ is contractible}\}.$$

These classes connect graph dismantlability [7] with non-evasiveness, collapsibility and contractibility of simplicial complexes [8, 9, 10]. Related graph models describe strong homotopy [11] and simple homotopy [12, 13]; collapsibility of flag complexes also arises in probabilistic settings [14]. Work on difficult triangulations and discrete Morse theory [15] provides context for the non-collapsible examples studied here. Small complexes with prescribed homology torsion [16] concern a broader class; the flag condition is essential to our vertex-minimality comparison.

Dochtermann, Espinoza, Frías-Armenta and Hernández [6] enumerated these graph classes through order 11, found the smallest members of $\mathfrak{I}_S \setminus \mathfrak{I}_{VS}$, and exhibited a 15-vertex flag triangulation $D_{15}$ of the dunce hat [17] in $\mathfrak{I} \setminus \mathfrak{I}_S$; their Fig. 2A shows $D_{15}$ and their Fig. 2B shows $D_{15}$ with one added edge. They stated the following conjectures [6, pp. 8–9]:

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**Conjecture 14.** *The smallest graph (in terms of vertices) with trivial $\mathbb{Z}_2$-homology that it is not strong $\mathfrak{I}$-contractible is shown in Fig. 2A. In particular this is the smallest graph that is $\mathfrak{I}$-contractible but not strong $\mathfrak{I}$-contractible.*

**Conjecture 15.** *The graph depicted in Fig. 2B is the smallest graph $G$ (in terms of vertices) that has the following properties:* (1) *$G$ is strong $\mathfrak{I}$-contractible;* (2) *There exists an edge $\{v, w\}$ in $G$ such that $N(v, w) \in \mathfrak{I}_S$ and such that $G \setminus \{v, w\}$ is not strong $\mathfrak{I}$-contractible.*

Thus Conjecture 14 asserts that $D_{15}$ has the fewest vertices among $\mathbb{Z}_2$-acyclic graphs outside $\mathfrak{I}_S$, and in particular among graphs in $\mathfrak{I} \setminus \mathfrak{I}_S$. Conjecture 15 asserts that $D_{15}$ with the extra edge of their Fig. 2B is a smallest graph $G \in \mathfrak{I}_S$ with an edge $uv$ such that $N_G(u, v) \in \mathfrak{I}_S$ but $G - uv \notin \mathfrak{I}_S$.[1]

The examples in this paper refute both conjectures. The two assertions of Conjecture 14 fail for different reasons: odd torsion is invisible to $\mathbb{Z}_2$-homology, whereas the gap $\mathfrak{I} \setminus \mathfrak{I}_S$ requires contractible clique complexes. We separate the directly checked examples and finite certificates from the completeness assertions supported by the large enumerations.

Main results.

(A) *Odd torsion on thirteen vertices.* Three explicit $\mathbb{Z}_2$-acyclic graphs outside $\mathfrak{I}_S$ have clique complexes homotopy equivalent to $M(\mathbb{Z}/3, 1)$. Consequently none belongs to $\mathfrak{I}$. Together with Adamaszek's classification [19], they give the minimum order 13 for odd torsion in $H_1$ of a flag complex (Remark 6.2). Under the computational hypotheses of Section 3, these are precisely the $\mathbb{Z}_2$-acyclic graphs outside $\mathfrak{I}_S$ on at most 13 vertices (Theorem 6.1).

(B) *The gap on fourteen vertices.* The three $K_4$-free graphs $G_{14}^a$, $G_{14}^b$, $G_{14}^c$ belong to $\mathfrak{I} \setminus \mathfrak{I}_S$. Each admits an (I4) edge addition followed by a verified strong deletion sequence. The recorded search returns 51 stuck graphs, of which the other 48 have $H_1 \cong \mathbb{Z}/3$. Under the stated computational hypotheses, 14 is the minimum order in $\mathfrak{I} \setminus \mathfrak{I}_S$ and exactly eight $K_4$-free graphs of that order lie in the gap (Theorem 6.4).

(C) *Reduction.* A graph of minimum order in either class is obtained from a stuck graph by admissible edge additions. The cut-vertex and degree lemmas reduce the search to 2-connected graphs with $4 \leq \delta \leq \Delta \leq n - 2$, using the reported exclusion of small almost stuck graphs (Theorem 3.7).

(D) *Failure of a legal strong edge deletion.* The three one-edge extensions in (B) refute Conjecture 15. Its minimum order is 14 under the same hypotheses; an independently recomputed finite search provides at least 32 non-isomorphic examples in the strong boundary of the pruned closure of $G_{14}^a$ (Theorem 7.1).

(E) *Cardinalities.* Table 2 reproduces the published rows through order 11 and adds rows for orders 12 and 13 with every column filled, including $|\mathfrak{I}_{VS}(12)| = 22{,}666{,}525{,}713$ and $|\mathfrak{I}_{VS}(13)| = 5{,}619{,}007{,}241{,}580$. These two rows are reported as computed: their strong-membership entries carry the implementation qualification of Section 4, and their completeness rests on the integrity of the recorded runs (Appendix D), which planned recomputations will check. The $K_4$-free order-14 row remains a *computation in progress*.

Reproducibility and scope. Data, code and certificates are in the public repository

https://github.com/Geometric-and-Combinatorial-Structures/minimal-I-contractible-not-strong

[1] Both conjectures are restated in the final section of [6] (p. 13), Conjecture 14 there without its second sentence. The preprint version [18] states Conjecture 14 for graphs "with trivial homology" under the convention $H_n(G) := H_n(G; \mathbb{Z})$. With that reading the first assertion concerns $\mathbb{Z}$-acyclic graphs outside $\mathfrak{I}_S$; it is refuted as well, by the contractible 14-vertex graphs of Theorem 6.4. Under the computational hypotheses the minimum order for that reading is 14 rather than 13, because the torsion graphs of Theorem 6.1 are not $\mathbb{Z}$-acyclic.

(Section 8). Independent Python checks verify the listed examples, integer homology and strong deletion certificates; Wolfram Language independently verifies the homology of the seven principal graphs. These finite checks do not repeat the large searches. The archived C implementation uses hash-only memoisation for strong membership. In the order-14 search, the copy of one worker script archived with the run was misidentified; the script that ran is identified by forensic corroboration from file-system records and task logs, not by an authenticated copy of the submitted script (Appendix D); and the two search fronts wrote to shared output paths. Section 4 and Appendix D distinguish the verified records from the checks still required for an unconditional computational certification of all minimality and completeness assertions.

Organisation. Sections 2 and 3 give the definitions and reduction arguments. Section 4 explains the algorithms and their validation; Table 2 supplies the small-order computational input before the examples of Section 6. Section 7 discusses consequences and open questions. The appendices contain the reduction proofs, the graph list, finite certificates and cluster records.

## 2. Preliminaries

All graphs are finite and simple. For a graph $G$ and a vertex $v$ we write $N(v) = N_G(v)$ for the *open* neighbourhood (the subgraph induced by the vertices adjacent to $v$), and for distinct vertices $u, v$ we write $N(u, v) = N(u) \cap N(v)$ for the *common neighbourhood*, again as an induced subgraph. $G - v$ and $G - uv$ denote deletion of a vertex and of an edge; $G + uv$ denotes the addition of an edge between non-adjacent vertices. The *clique complex* $\Delta(G)$ is the simplicial complex whose simplices are the cliques of $G$; a complex of this form is called *flag*. Note that $\Delta(N(v))$ is the link of $v$ in $\Delta(G)$ and $\Delta(N(u, v))$ is the link of $uv$ when $uv$ is an edge.

### 2.1. Contractible transformations

We follow the conventions of [6].

**Definition 2.1** ([2])**.** The class $\mathfrak{I}$ of *I-contractible* graphs is the smallest class of graphs containing $K_1$ and closed under the following four transformations:

(I1) deleting a vertex $v$ with $N(v) \in \mathfrak{I}$;
(I2) adding a new vertex $v$ adjacent to the vertices of an induced subgraph $H \in \mathfrak{I}$;
(I3) deleting an edge $uv$ with $N(u, v) \in \mathfrak{I}$;
(I4) adding an edge $uv$ between non-adjacent vertices with $N(u, v) \in \mathfrak{I}$.

The four transformations are the *contractible transformations* of Ivashchenko; (I3) and (I4) can be realised by (I1) and (I2) [6, Lemma 3]. Chen, Yau and Yeh [4] proved that these transformations preserve the simple homotopy type of the clique complex. Hence every graph in $\mathfrak{I}$ has contractible $\Delta(G)$; the converse is [5, Corollary 1.5].

**Definition 2.2** ([6])**.** The class $\mathfrak{I}_S$ of *strong I-contractible* graphs is defined recursively as follows. The graph $K_1$ belongs to $\mathfrak{I}_S$; a graph $G$ with at least two vertices belongs to $\mathfrak{I}_S$ if at least one of the following conditions holds:

(S1) $G - v \in \mathfrak{I}_S$ and $N_G(v) \in \mathfrak{I}_S$ for some vertex $v$;
(S2) $G - uv \in \mathfrak{I}_S$ and $N_G(u, v) \in \mathfrak{I}_S$ for some edge $uv$.

The empty graph does not belong to $\mathfrak{I}_S$. Equivalently, $G \in \mathfrak{I}_S$ if it has a sequence of these deletions ending at $K_1$. The class $\mathfrak{I}_{VS}$ of *vertex-strong I-contractible* graphs is defined by the same recursion using only vertex deletions and requiring both $G - v$ and $N_G(v)$ to belong to $\mathfrak{I}_{VS}$.

The empty graph is *not* in $\mathfrak{I}_S$; in particular an edge $uv$ with $N(u, v) = \emptyset$ is never deleted, and an isolated vertex is never deleted. A vertex $v$ with $N(v) \in \mathfrak{I}_S$, or an edge $uv$ with $N(u, v) \in \mathfrak{I}_S$, will be called $\mathfrak{I}_S$*-deletable*. In a graph of $\mathfrak{I}_S$ the order of deletions matters: a sequence may lead to a graph with no $\mathfrak{I}_S$-deletable element while another sequence reaches $K_1$ ([6, Fig. 2B]); thus a greedy procedure does not decide $\mathfrak{I}_S$ and an exact search is needed (Section 4).

**Definition 2.3** ([6, 7])**.** A vertex $v$ of $G$ is 0-*dismantlable* if $N(v)$ is a cone (some vertex of $N(v)$ is adjacent to all others), and, for $k \geq 1$, $k$-*dismantlable* if $N(v)$ is $(k-1)$-dismantlable; $(-1)$-dismantlable graphs are the cones. A graph is $k$-*dismantlable* ($k$-Dis) if it can be reduced to $K_1$ by deleting $k$-dismantlable vertices. A 0-dismantlable graph is also called *dismantlable* or *cop-win* [20].

One has $(-1)$-Dis $\subseteq$ 0-Dis $\subseteq$ 1-Dis $\subseteq \cdots \subseteq \mathfrak{I}_{VS} \subseteq \mathfrak{I}_S \subseteq \mathfrak{I}$ ([6, Lemma 19] and induction).

### 2.2. **Homology**

We write $\widetilde{H}_i(G; R)$ for the reduced simplicial homology of $\Delta(G)$ with coefficients in a ring $R$, and $\chi(G)$ for the Euler characteristic of $\Delta(G)$, which can be computed from the clique numbers $c_k(G)$ as $\chi(G) = \sum_{k\geq 1}(-1)^{k-1}c_k(G)$. A graph is $\mathbb{Z}_2$-*acyclic* if it is non-empty and $\widetilde{H}_i(G; \mathbb{Z}_2) = 0$ for every $i \geq 0$; in particular it is connected. Define $\mathbb{Z}$-*acyclicity* similarly. The universal coefficient theorem implies that $\mathbb{Z}$-acyclicity implies $\mathbb{Z}_2$-acyclicity. For a finite connected complex, $\mathbb{Z}_2$-acyclicity is equivalent to all its positive-degree integral homology groups being finite of odd order; integral acyclicity additionally requires these groups to vanish. A simplicial complex is *contractible* if it is homotopy equivalent to a point. Contractibility implies integral acyclicity, but the converse is false in general: an acyclic complex can have a non-trivial perfect fundamental group. We establish contractibility of our examples by explicit I-transformations, rather than inferring it from integral homology alone.

Integer homology of the small complexes in this paper is computed by exact Smith normal form of the boundary matrices (Section 4). We use the following elementary fact repeatedly.

**Lemma 2.4.** *Let $n$ be the smallest order of a $\mathbb{Z}_2$-acyclic graph that is not in $\mathfrak{I}_S$ (if such graphs exist). Then for every graph $H$ with fewer than $n$ vertices, $H \in \mathfrak{I}_S$ if and only if $H$ is $\mathbb{Z}_2$-acyclic.*

*Proof.* Graphs in $\mathfrak{I}_S$ are in $\mathfrak{I}$, hence have contractible clique complex and are $\mathbb{Z}_2$-acyclic; the converse holds below $n$ by minimality of $n$. □

### 2.3. **Discs with identified boundary**

For the structural descriptions in Section 6, let $X$ be a pure 2-dimensional simplicial complex. Start with disjoint copies of its triangles and glue corresponding edges only when the original edge belongs to exactly two triangles. Identify the corresponding corners under these gluings. This produces a complex $\widetilde{X}$ with a natural map to $X$; in each case discussed below, we check that $\widetilde{X}$ is a compact surface with boundary. Thus the construction cuts along every edge whose triangle incidence differs from two, including edges in four triangles. Purity alone does not guarantee that the result is a surface. For the stuck $K_4$-free examples every edge belongs to at least two triangles, so the cut occurs precisely along the singular edges.

Suppose that $\widetilde{X}$ is a disc, the map is injective on its interior, and $X$ is recovered solely by identifying boundary points according to their image. Let $S$ be the image of the boundary, and suppose that $S$ is a connected graph with exactly one cycle $C$. Then $X$ is the adjunction space $S \cup_f D^2$, where the boundary word records the attaching map $f$. Contract the trees attached to $C$ and orient $C$. If the resulting boundary map has degree $d$, then

$$X \simeq S^1 \cup_d D^2.$$

Indeed, collapsing a maximal tree in $S$ is a homotopy equivalence of the adjunction spaces, and maps $S^1 \to S^1$ are classified up to homotopy by their degree. In particular, $|d| = 1$ gives a contractible complex, while $|d| = 3$ gives the Moore space $M(\mathbb{Z}/3, 1)$ up to homotopy. When the boundary makes three complete passes around $C$, the patterns $a\,a\,a^{-1}$ and $a^3$ have degrees 1 and 3, respectively. Excursions into an attached tree contribute degree zero.

These hypotheses are verified for $G_{13}$, the 15-vertex dunce hat $D_{15}$ of [6], and 48 of the 51 stuck $K_4$-free graphs on 14 vertices. Among the latter, three have degree $\pm 1$ and 45 have degree $\pm 3$. The

remaining three unfold into two annuli and one disconnected surface; the single-disc argument does not apply to them.

## 3. Reduction to stuck graphs

The number of connected graphs on 13 vertices (50,335,907,869,219 $\approx 5.0 \times 10^{13}$) and on 14 vertices ($\approx 2.9 \times 10^{16}$; OEIS A001349) makes a naive search for the smallest member of $\mathfrak{I} \setminus \mathfrak{I}_S$ impractical: each graph would also require a strong-membership decision. The following observation turns the search for a smallest counterexample into a search for a much rarer kind of graph. Computational hypotheses. To distinguish formal reductions from completeness claims depending on the recorded runs, we make the following assumptions explicit:

H12. The classifications in Table 2 through order 12 agree with complete evaluations of the definitions, with full graph states compared whenever memoisation is used.

Hcasi. The recorded `casi` runs exhaust all connected graphs of orders 3 through 11 and return no graph.

H13. The run `geng -C -d4 -D11 13` exhausts the specified class, and applying the homology filter to it gives precisely the single graph in `data/acyclic_not_strong_13.g6` that has 40 edges.

H14. The run `geng -k -C -d4 -D12 14` exhausts the specified class, and applying the homology filter gives precisely the 51 graphs in `data/stuck_14_K4free.g6`.

The logs support these assumptions, but the audit identified limitations of the table implementation and the cluster records, discussed in Section 4. Completeness and minimality statements below are conditional on the indicated assumptions. Finite checks of the displayed examples and their closures are separate from these large enumerations.

**Definition 3.1.** A graph $G$ with at least two vertices is *stuck* if it is $\mathbb{Z}_2$-acyclic and has no $\mathfrak{I}_S$-deletable vertex and no $\mathfrak{I}_S$-deletable edge, i.e. $N(v) \notin \mathfrak{I}_S$ for every vertex $v$ and $N(u,v) \notin \mathfrak{I}_S$ for every edge $uv$ (recall that $\emptyset \notin \mathfrak{I}_S$). A graph $A$ with at least two vertices and a distinguished vertex $u$ is *almost stuck at* $u$ if it is $\mathbb{Z}_2$-acyclic, has no $\mathfrak{I}_S$-deletable edge, and no vertex other than $u$ is $\mathfrak{I}_S$-deletable ($u$ itself may or may not be).

A stuck graph is trivially outside $\mathfrak{I}_S$: it admits no strong contractible move at all. Conversely:

**Lemma 3.2** (Minimality implies stuck)**.** *(a) Let $n_0$ be the smallest order of a $\mathbb{Z}_2$-acyclic graph not in $\mathfrak{I}_S$. Every $\mathbb{Z}_2$-acyclic graph on $n_0$ vertices not in $\mathfrak{I}_S$ is obtained from a stuck graph on $n_0$ vertices by successively adding edges $uv$ with $N(u,v) \in \mathfrak{I}_S$ (transformations (I4)); we say it lies in the* $\mathrm{I}_4$-closure *of that stuck graph. The closure includes the starting graph. Those with the minimum number of edges are themselves stuck.*

*(b) Let $n_1$ be the smallest order of a graph in $\mathfrak{I} \setminus \mathfrak{I}_S$. Every graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ on $n_1$ vertices lies in the $\mathrm{I}_4$-closure of a stuck graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ on $n_1$ vertices, which is $K_4$-free if the original graph is. Those with the minimum number of edges (in the whole class, or in the $K_4$-free class) are stuck.*

**Lemma 3.3** (Stuck graphs are 2-connected)**.** *Let $G$ be a stuck graph with a cut vertex $u$, so that $G = A \cup_u B$ with $A$, $B$ induced and connected, $|V(A)|, |V(B)| \geq 2$, $V(A) \cap V(B) = \{u\}$ and no edges between $A - u$ and $B - u$. Then $A$ and $B$ are almost stuck at $u$ and have at least three vertices. Consequently, if there is no almost stuck graph with at most $m$ vertices, every stuck graph with at most $2m$ vertices is 2-connected.*

**Lemma 3.4** (Degree bounds)**.** *Let $G$ be a 2-connected stuck graph on $n \geq 2$ vertices. Then every edge of $G$ lies in a triangle, every common neighbourhood $N(u,v)$ of the ends of an edge is non-empty, $\delta(G) \geq 4$ and $\Delta(G) \leq n - 2$.*

The proofs are elementary and are given in Appendix A; Fig. 1 illustrates Lemma 3.3.

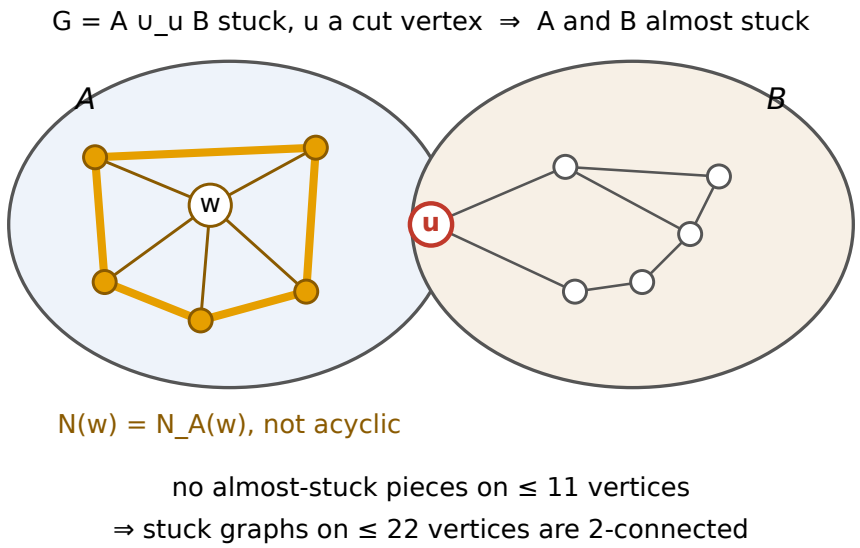



FIGURE 1. Lemma 3.3: if a stuck graph $G = A \cup_u B$ has a cut vertex $u$, the neighbourhoods of all vertices and edges of $A$ other than $u$ are computed inside $A$, so $A$ is almost stuck at $u$. Under Hcasi, Proposition 3.6 excludes almost stuck graphs on at most 11 vertices and implies 2-connectedness through order 22. In the preserved graphic, "not acyclic" should be read as "not in $\mathfrak{I}_S$" for the general cut-vertex lemma; the homology replacement needs the small-neighbourhood hypothesis of Lemma 3.5.

To test stuckness during an enumeration one would like to replace the condition "$\notin \mathfrak{I}_S$" on neighbourhoods by the cheaper "not $\mathbb{Z}_2$-acyclic". This is legitimate in the range of this paper:

**Lemma 3.5.** *Suppose that every $\mathbb{Z}_2$-acyclic graph with at most $r$ vertices belongs to $\mathfrak{I}_S$. Let $G$ be a $\mathbb{Z}_2$-acyclic graph on $2 \le n \le r+2$ vertices without a universal vertex. Then $G$ is stuck if and only if no open neighbourhood and no non-empty common neighbourhood of an edge is $\mathbb{Z}_2$-acyclic. For $2 \le n \le r+1$, the analogous criterion for being almost stuck at a specified vertex $u$ holds without excluding universal vertices: no edge may have acyclic common neighbourhood, and no vertex other than $u$ may have acyclic open neighbourhood.*

*Proof.* Membership in $\mathfrak{I}_S$ always implies $\mathbb{Z}_2$-acyclicity. Under the hypothesis the converse holds for neighbourhoods of order at most $r$. If $G$ has no universal vertex, its open neighbourhoods have at most $n-2$ vertices; edge common neighbourhoods always have at most $n-2$. In the almost-stuck case an open neighbourhood has at most $n-1 \le r$ vertices even when its vertex is universal. Empty neighbourhoods belong to neither class. □

For the searches at $n = 13$ and 14 we use this lemma with $r = 12$, assuming H12. The cardinality computation uses the recursive $\mathfrak{I}_S$ decision, not the stuck filter. The logical order is: the table through order 12; the almost-stuck check through order 11; the reduction; the searches at orders 13 and 14. In particular the 14-vertex stuck test does not require the classification at order 13 in Theorem 6.1.

**Proposition 3.6.** *Assuming* Hcasi, *there is no almost stuck graph with at most* 11 *vertices. Consequently every stuck graph with at most* 22 *vertices is* 2*-connected.*

*Proof.* The only acyclic graph of order 2 is $K_2$, whose two vertices are deletable, so it is not almost stuck. An almost stuck graph of order at most 11 in which $u$ is not $\mathfrak{I}_S$-deletable would be stuck and hence outside $\mathfrak{I}_S$, contrary to the equality of the acyclic and $\mathfrak{I}_S$ columns in the published table through order 11 [6, Table 1]. If $u$ is deletable, it is the unique deletable vertex. The mode `casi` counts exactly one vertex with acyclic neighbourhood, rejects edges with acyclic common neighbourhood and tests acyclicity of the graph. By Lemma 3.5, applied with $r = 10$, these tests are equivalent to this remaining case, including graphs with universal vertices. Under Hcasi, the run on all connected graphs with $3 \leq n \leq 11$ found none (1,006,700,565 graphs at order 11). The second statement follows from Lemma 3.3: a cut vertex would give $|A|, |B| \geq 12$ and hence $|G| \geq 23$. □

**Theorem 3.7** (Reduction)**.** *(i) If $n_0 \leq 22$ is the smallest order of a $\mathbb{Z}_2$-acyclic graph outside $\mathfrak{I}_S$, every such graph on $n_0$ vertices lies in the $\mathrm{I}_4$-closure of a stuck graph on $n_0$ vertices.*

*(ii) If $n_1 \leq 22$ is the smallest order of a graph in $\mathfrak{I} \setminus \mathfrak{I}_S$, every graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ on $n_1$ vertices lies in the $\mathrm{I}_4$-closure of a stuck graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ on $n_1$ vertices, $K_4$-free if the graph is.*

*Assuming* Hcasi*, in both cases every stuck graph on $n_i$ vertices is 2-connected with $4 \leq \delta \leq \Delta \leq n_i - 2$. Assuming also* H12*, for $n_i \leq 14$ stuckness is decided by $\mathbb{Z}_2$-homology of the neighbourhoods (Lemma 3.5). Hence, to find all graphs of either kind on $n_i$ vertices it suffices to enumerate the 2-connected graphs with these degree bounds, keep the stuck ones, and compute their $\mathrm{I}_4$-closures, retaining the graphs in the class under consideration.*

For $n = 13$ the enumeration of 2-connected graphs with $4 \leq \delta$ and $\Delta \leq 11$ comprises 21,011,415,992,470 graphs, about 2.4 times fewer than all connected graphs; the real gain lies in the stuck filter: the cascade of Section 4 rejects all but a fraction $7.5 \times 10^{-4}$ of the graphs by looking at cones, the Euler characteristic and a few neighbourhoods, computes the homology of $G$ itself for 218,516 graphs after the edge-neighbourhood filter, and never runs the exact $\mathfrak{I}_S$ decision during the enumeration. For $n = 14$ we restricted the search to $K_4$-free graphs (clique complexes of dimension at most 2); this class contains the dunce hat $D_{15}$ and comprises 64,856,470,381,820 graphs under the same degree constraints. Together with H12, Hcasi, H13 and H14, Theorem 3.7(ii) gives completeness for the $K_4$-free members of $\mathfrak{I} \setminus \mathfrak{I}_S$ on 14 vertices.

## 4. Algorithms and the computation on YUCA

The graph-search routines are implemented in a C program (`src/atascadas.c` in the repository; *atascada* is Spanish for *stuck*) that reads graphs in `graph6` format from the output of `geng` [21]. Graphs have at most 15 vertices, so vertex sets are bitmasks and neighbourhoods are arrays of bitmasks; cliques are enumerated by recursive extension. Independent re-implementations in pure Python of the exact $\mathfrak{I}_S$ decision and of integer homology (`tools/`) are used for the certificates (Section 8).

### 4.1. Homology and the stuck filter

$\mathbb{Z}_2$-homology of $\Delta(H)$ for a small graph $H$ is computed by Gaussian elimination over GF(2) of the boundary matrices, with simplices up to dimension $\omega(H) - 1$. The Euler characteristic $\chi(H) = \sum_k (-1)^{k-1} c_k(H)$ is obtained from the clique enumeration and is a much cheaper necessary condition ($\chi = 1$) for $\mathbb{Z}_2$-acyclicity. Integer homology of the candidates is computed by exact Smith normal forms over $\mathbb{Z}$. For the 51 stuck graphs on 14 vertices, the largest edge–triangle boundary matrix is $47 \times 34$. The torsion of $H_k$ equals that of the cokernel of $\partial_{k+1}$, since the quotient by $H_k$ is the free abelian group $\operatorname{im} \partial_k$; the free rank is obtained from the two adjacent boundary ranks.

Under H12 of Section 3, Algorithm 1 decides whether a graph in the stated search range is stuck by a cascade of tests of increasing cost; the tests are ordered so that the overwhelming majority of graphs is rejected early.

**Algorithm 1:** STUCK$(G)$ — filter cascade

**Input:** a connected graph $G$ with $2 \le n \le 14$, no universal vertex, and hypothesis H12
**Output:** **true** iff $G$ is stuck

**if** $|N(v)| < 2$ ***or*** $N(v)$ *is a cone* **then return false**;
**if** $\chi(G) \neq 1$ **then return false**;
**foreach** *vertex* $v$ **do**
  **if** $N(v)$ *is* $\mathbb{Z}_2$*-acyclic* **then return false**;
**foreach** *edge* $uv$ **do**
  **if** $N(u,v) \neq \emptyset$ ***and*** $N(u,v)$ *is* $\mathbb{Z}_2$*-acyclic* **then return false**;
**if** $G$ *is not* $\mathbb{Z}_2$*-acyclic* **then return false**;
**return true**

**Algorithm 2:** INIS$(G)$ — exact membership in $\mathfrak{I}_S$

**Input:** a graph $G = (V, E)$
**Output:** **true** iff $G \in \mathfrak{I}_S$ (and, on success, a deletion sequence)

**if** $|V| = 0$ **then return false**;
**if** $|V| = 1$ ***or*** $G$ *is* 0*-dismantlable* **then return true**;
**if** $(V, E)$ *is memoised* **then return** the stored answer;
**foreach** *vertex* $v$ *with* $N(v) \neq \emptyset$ *and* INIS$(N(v))$ **do**
  **if** INIS$(G - v)$ **then** memoise and **return true**;
**foreach** *edge* $uv$ *with* $N(u,v) \neq \emptyset$ *and* INIS$(N(u,v))$ **do**
  **if** INIS$(G - uv)$ **then** memoise and **return true**;
memoise and **return false**

The acyclicity test for a neighbourhood $H$ is itself a cascade: $H$ disconnected $\Rightarrow$ not acyclic; $H$ 0-dismantlable (greedy cone-vertex removal) $\Rightarrow$ acyclic; $\chi(H) \neq 1 \Rightarrow$ not acyclic; otherwise compute $\widetilde{H}_*(H; \mathbb{Z}_2)$. The *almost stuck* variant used for Proposition 3.6 omits the initial cone/degree rejection and permits universal vertices. It tests $\chi = 1$, counts vertices with acyclic neighbourhood and accepts exactly one, then applies the edge-neighbourhood and full homology tests.

### 4.2. Exact decision of $\mathfrak{I}_S$, $\mathfrak{I}_{VS}$ and $k$-Dis

Since the one-edge certificates of Section 6.2 show that a legal strong edge deletion can leave $\mathfrak{I}_S$, we decide $\mathfrak{I}_S$ by exhaustive search over deletion sequences (Algorithm 2). The mathematical algorithm memoises the complete pair (vertex set, edge set) of the current graph; a 0-dismantlable graph is accepted immediately ([6, Lemma 19]); the neighbourhood conditions are decided recursively by the same procedure. $\mathfrak{I}_{VS}$ and $k$-Dis are decided by dynamic programming over the $2^n$ induced subgraphs: a set $S$ is reducible if $|S| = 1$ or some $v \in S$ has $N_{G[S]}(v)$ in the relevant class and $S \setminus \{v\}$ reducible.

Implementation qualification. The Python implementation stores complete graph states and implements Algorithm 2 exactly. The archived C versions instead use a 64-bit hash as the identity of a memoised state, without a final comparison of the underlying graph. The audit did not demonstrate an actual collision, but the implementation alone does not certify its absence. This affects the strong-membership part of the order-12 table, which supplies hypothesis H12. Its acyclicity and vertex-strong decisions use exact arithmetic and subset-indexed states. A concrete way to close the

**Algorithm 3:** ICERT$(G, d)$ — certificate of I-contractibility by $\leq d$ additions

$Q \leftarrow \{(G, \emptyset)\}$;
**for** $\ell = 0, \ldots, d$ **do**
  **foreach** $(H, L) \in Q$ **do**
    **if** INIS$(H)$ **then return** $(L,$ deletion sequence of $H)$;
  $Q \leftarrow \{(H + uv,\ L \cdot uv) : (H, L) \in Q,\ uv \notin E(H),\ N_H(u, v) \in \mathfrak{I}_S\}$, one representative per isomorphism class;
**return undecided**

gap is to export the 11,718 acyclic, non-vertex-strong candidates and check their strong membership with complete-state memoisation. The large stuck searches use homology tests rather than this strong-membership cache, but interpreting their outputs relies on H12.

### 4.3. Certificates of I-contractibility and $I_4$-closures

To certify that a stuck graph $G$ is I-contractible we search breadth-first for a short sequence of edge additions $uv$ with $N(u, v) \in \mathfrak{I}_S$ (transformations (I4), legitimate since $\mathfrak{I}_S \subseteq \mathfrak{I}$) leading to a graph in $\mathfrak{I}_S$ (Algorithm 3); the certificate consists of the added edges together with the deletion sequence of the final graph produced by Algorithm 2, and can be checked by hand or by the Python tools. For all three examples on 14 vertices a single edge suffices.

To enumerate the part of an $I_4$-closure outside $\mathfrak{I}_S$, we expand only states outside $\mathfrak{I}_S$ and record their first successors in $\mathfrak{I}_S$. This pruning is valid because an admissible edge addition to a graph in $\mathfrak{I}_S$ remains in $\mathfrak{I}_S$: its inverse deletion starts a successful reduction. We call the retained non-strong states the *pruned closure* and the first strong successors its *strong boundary*. The C program deduplicates labelled states by hashes; canonical labelling with `labelg` is a separate step. The finite closures used here were also recomputed with complete-state Python storage. For the 51 listed stuck graphs on 14 vertices this produces the $\mathbb{Z}_2$-acyclic graphs outside $\mathfrak{I}_S$ reached from those seeds (Section 6.2).

### 4.4. Unfolding into a disc

For a pure two-dimensional candidate we retain the gluing across edges in exactly two triangles and cut across all other edges, splitting vertices according to the resulting corner equivalence (Section 2); when the result is a disc we draw it with a Tutte embedding (boundary on a regular polygon, interior vertices at the barycentres of their neighbours) and read off the boundary word. The resulting complexes must be checked for connectedness, vertex links, boundary components and Euler characteristic before they are called discs or annuli. This procedure yields 48 discs, two annuli, and one disjoint union of a disc and an annulus among the 51 seeds.

### 4.5. The distributed search

The enumeration used the CPU partition of the YUCA cluster at ACARUS, Universidad de Sonora, with 64 cores per node and `geng`'s `res/mod` partitioning. The order-14 search used `nauty` 2.8 for its $K_4$-free option `-k`. Figure 2 summarises the mathematical pipeline; Appendix D gives the scheduling procedure, hardware and provenance qualifications.

For $n = 14$ two projects processed slices in opposite orders. The reconstructed registries contain 2,000,692 entries, covering every one of the 2,000,000 slices, with 692 slices recorded twice. The counters in Table 1 sum the per-slice files once. The two workers used different lock files while writing shared result paths, so the surviving archive does not establish that both executions of each duplicated slice produced identical output. Repeating those slices with separate output files remains a certification requirement for H14.

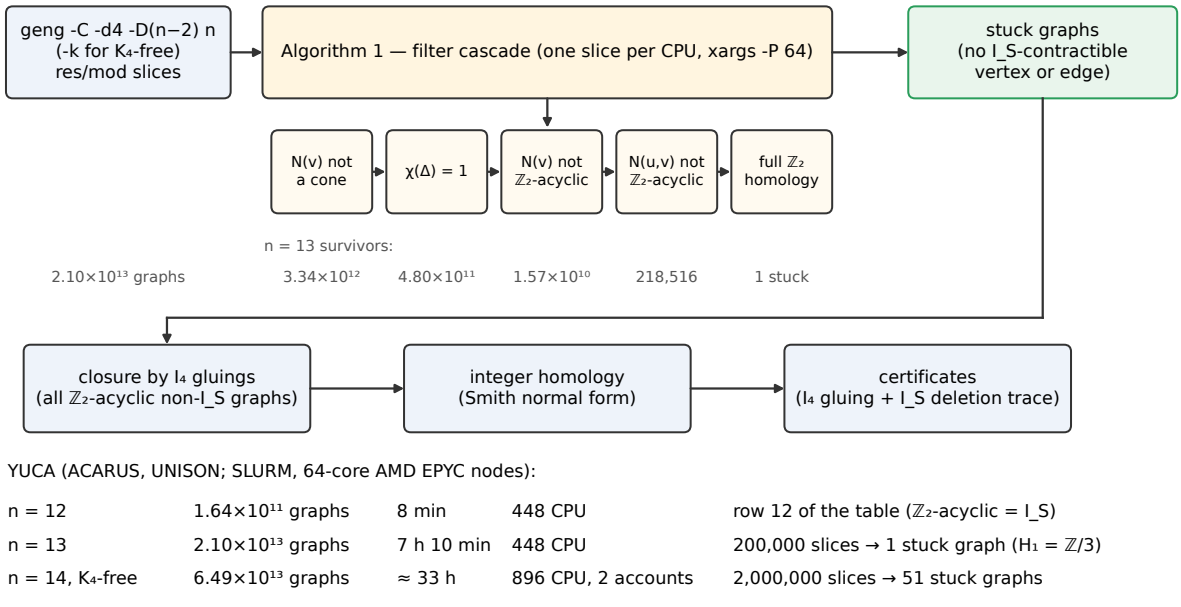



FIGURE 2. The computational pipeline: `geng` enumeration in `res/mod` slices, the filter cascade of Algorithm 1 (with the survivor counts of the $n = 13$ run), $I_4$-closures, integer homology and certificates; bottom: the three cluster runs. The phrase "all $\mathbb{Z}_2$-acyclic non-$\mathfrak{I}_S$ graphs" applies to order 13 under the stated hypotheses; at order 14 only descendants of the listed seeds are covered. The preserved graphic shows the earlier approximate duration 33 h; the accounting gives 35 h 2 min 48 s for the order-14 array span.

TABLE 1. Searches performed. "graphs" is the number of graphs produced by `geng` and examined; the filter counters are the numbers of graphs passing the cone test, the Euler characteristic test and the neighbourhood tests of Algorithm 1.

| $n$ | class (`geng` options) | mode | graphs | pass cone | pass $\chi$ | pass $N$ | stuck |
|---|---|---|---|---|---|---|---|
| 12 | connected (`-c`) | table | 164,059,830,476 | – | – | – | 0 |
| 13 | `-C -d4 -D11` | stuck | 21,011,415,992,470 | 3,335,926,431,910 | 480,204,455,162 | 15,693,920,490 | 1 |
| 14 | `-k -C -d4 -D12` | stuck | 64,856,470,381,820 | 48,303,369,131,820 | 1,462,235,073,536 | 108,022,349,478 | 51 |

| $n$ | slices $M$ | max. cores | wall time | SLURM arrays |
|---|---|---|---|---|
| 12 | 4,000 | 448 | 7 min 10 s | 2079049, 2079051 |
| 13 | 200,000 | 448 | 7 h 10 min | 2079058, 2079059 |
| 14 | 2,000,000 | 896 | 35 h 2 min 48 s | 2079102 (p029), 2079156 (p011) |

### 4.6. **Validation**

All 63 cells common to Table 2 and Table 1 of [6] agree (nine rows and seven columns). The script `verify.sh` of the repository passes all 39 of its checks when `gcc` and `geng` are present (16 October 2026), including rows regenerated through order 9. An external audit of commit `4a059b5` (15–16 October 2026) reproduced rows 3–9 graph by graph with independently written code, the complete row 10 with the archived C program and the complete row 11 (1,006,700,565 connected graphs) with a variant whose memo compares full graph states; all agree with Table 2. Rows 12

and 13 were not re-enumerated there. The historical order-12 comparison with the unpublished 2025 draft is limited to cells actually filled in that draft; the $\mathfrak{I}_{VS}$ cell there is blank.

The seven principal graphs were independently checked in Wolfram Language using integral kernel bases and Smith normal forms. The Python tools verify the homology and strong-membership status of all 51 seeds, the one-edge certificates and the cone routes. Complete-state closure enumeration and canonical labelling reproduce the finite closure counts. These checks establish the listed finite results; they do not replace the computational hypotheses needed for exhaustivity of the original searches.

## 5. The extended cardinality table

Table 2 reports the numbers of connected graphs in the classes of Section 2. The seven shared columns in rows $3 \le n \le 11$ agree with Table 1 of [6], which already includes $\mathfrak{I}$. For $n \le 12$, the reported equality of the acyclic and strong columns gives the $\mathfrak{I}$ column by the inclusions $\mathfrak{I}_S \subseteq \mathfrak{I} \subseteq \{\mathbb{Z}_2\text{-acyclic graphs}\}$; no separate search for arbitrary I-transformations is required. The order-12 row combines two runs: the first computed six columns, including $|\mathfrak{I}_{VS}(12)| = 22{,}666{,}525{,}713$; the second, with the final program, computed all columns and reproduced those six exactly. The order-13 row was computed in 200,000 slices of `geng -c 13` (Appendix D); its connected counters sum to 50,335,907,869,219, the number of connected graphs on 13 vertices (OEIS A001349). This agreement is a consistency check; the further premises under which it would exclude missing or truncated slices are stated in Appendix D. Without the reductions of Section 3, the recorded run returns exactly three $\mathbb{Z}_2$-acyclic graphs outside $\mathfrak{I}_S$ without a certificate of at most three (I4) additions, and these are the three torsion graphs of Theorem 6.1; hence the reported row has $|\mathfrak{I}(13)| = |\mathfrak{I}_S(13)|$ and agrees with Theorem 6.1 by a route independent of the reduction. Rows 12 and 13 are reported as computed: hypothesis H12 and the implementation qualification in Section 4 apply to their strong-membership entries, and the completeness of both rows rests on the integrity of the recorded runs (Appendix D), which planned recomputations will check.

Several patterns in the table are worth recording.

*Observation* 5.1.

(i) Under H12, for $n \le 12$ every $\mathbb{Z}_2$-acyclic graph is strong I-contractible: $\mathbb{Z}_2$-acyclic $= \mathfrak{I} = \mathfrak{I}_S$; in particular $\mathbb{Z}_2$- and $\mathbb{Z}$-acyclicity coincide for $n \le 12$. The first exceptions occur at $n = 13$ and are the three torsion graphs of Theorem 6.1; the first members of $\mathfrak{I} \setminus \mathfrak{I}_S$ occur at $n = 14$ (Theorem 6.4).

(ii) In the reported rows, $\mathfrak{I}_{VS} = 2\text{-Dis} = 3\text{-Dis}$ for all $n \le 13$. Hence, if the rows of orders 12 and 13 are complete, a graph in $\mathfrak{I}_{VS}$ that is not 2-dismantlable has at least 14 vertices; such graphs are known to exist [7, Proposition 8]. The $\mathfrak{I}_{VS}$ and $k$-Dis columns are decided by exact dynamic programming over vertex subsets and do not depend on the memoisation discussed in Section 4; the bound depends only on the completeness and integrity of the recorded runs (Appendix D). At $n = 12$ the equality agrees with an independent (unpublished, 2025) computation of $|2\text{-Dis}(12)|$ by the authors. On the other hand $1\text{-Dis} = 2\text{-Dis}$ for $n \le 11$, while $|2\text{-Dis}(12)| - |1\text{-Dis}(12)| = 20$ and $|2\text{-Dis}(13)| - |1\text{-Dis}(13)| = 35{,}757$; the twenty graphs of order 12 are listed in Table 5 (Appendix B.2), and their 2-dismantling sequences are in the repository (Section 8).

(iii) $\mathfrak{I}_S \neq \mathfrak{I}_{VS}$ first at $n = 11$ (twelve graphs; the smallest members of $\mathfrak{I}_S \setminus \mathfrak{I}_{VS}$ of [6]); the reported differences are $|\mathfrak{I}_S(12)| - |\mathfrak{I}_{VS}(12)| = 11{,}718$ and $|\mathfrak{I}_S(13)| - |\mathfrak{I}_{VS}(13)| = 14{,}948{,}703$.

(iv) $\chi = 1$ but not $\mathbb{Z}_2$-acyclic first at $n = 7$ (one graph), and the gap grows to 9,816,037,137 graphs at $n = 12$ and 4,240,933,193,936 at $n = 13$. The column $(-1)$-Dis counts cones, that is, all graphs on $n - 1$ vertices (OEIS A000088: 1,018,997,864 for $n = 12$ and 165,091,172,592 for $n = 13$), which provides a consistency check.

TABLE 2. Numbers of connected graphs on $n$ vertices in each class. Shared columns for $3 \le n \le 11$ agree with [6, Table 1]; rows 12 and 13 are new and are reported as computed by the recorded runs (Appendix D). †For $n = 14$ only the $K_4$-free class has been searched. Classification assertions at orders 13 and 14 use the computational hypotheses of Section 3; the strong-membership entries of rows 12 and 13 carry the qualification of Section 4.

| $n$ | connected | $\chi = 1$ | $\mathbb{Z}_2$-acyclic | $\mathfrak{I}$ | $\mathfrak{I}_S$ | $\mathfrak{I}_{VS}$ | 3-Dis | 2-Dis | 1-Dis | 0-Dis | $(-1)$-Dis |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 4 | 6 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 |
| 5 | 21 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 16 | 11 |
| 6 | 112 | 68 | 68 | 68 | 68 | 68 | 68 | 68 | 68 | 68 | 34 |
| 7 | 853 | 404 | 403 | 403 | 403 | 403 | 403 | 403 | 403 | 403 | 156 |
| 8 | 11,117 | 3,837 | 3,798 | 3,798 | 3,798 | 3,798 | 3,798 | 3,798 | 3,798 | 3,791 | 1,044 |
| 9 | 261,080 | 68,458 | 66,167 | 66,167 | 66,167 | 66,167 | 66,167 | 66,167 | 66,167 | 65,561 | 12,346 |
| 10 | 11,716,571 | 2,571,263 | 2,343,148 | 2,343,148 | 2,343,148 | 2,343,148 | 2,343,148 | 2,343,148 | 2,343,148 | 2,258,314 | 274,668 |
| 11 | 1,006,700,565 | 204,131,202 | 166,978,401 | 166,978,401 | 166,978,401 | 166,978,389 | 166,978,389 | 166,978,389 | 166,978,389 | 151,084,672 | 12,005,168 |
| 12 | 164,059,830,476 | 32,482,574,568 | 22,666,537,431 | 22,666,537,431 | 22,666,537,431 | 22,666,525,713 | 22,666,525,713 | 22,666,525,713 | 22,666,525,693 | 18,510,237,804 | 1,018,997,864 |
| 13 | 50,335,907,869,219 | 9,859,955,384,222 | 5,619,022,190,286 | 5,619,022,190,283 | 5,619,022,190,283 | 5,619,007,241,580 | 5,619,007,241,580 | 5,619,007,241,580 | 5,619,007,205,823 | 4,015,494,549,880 | 165,091,172,592 |
| 14† | | *computation in progress*; stuck $\mathbb{Z}_2$-acyclic $\setminus \mathfrak{I}_S = 51$, $\mathfrak{I} \setminus \mathfrak{I}_S = 8$ (conditional; see text) | | | | | | | | | |

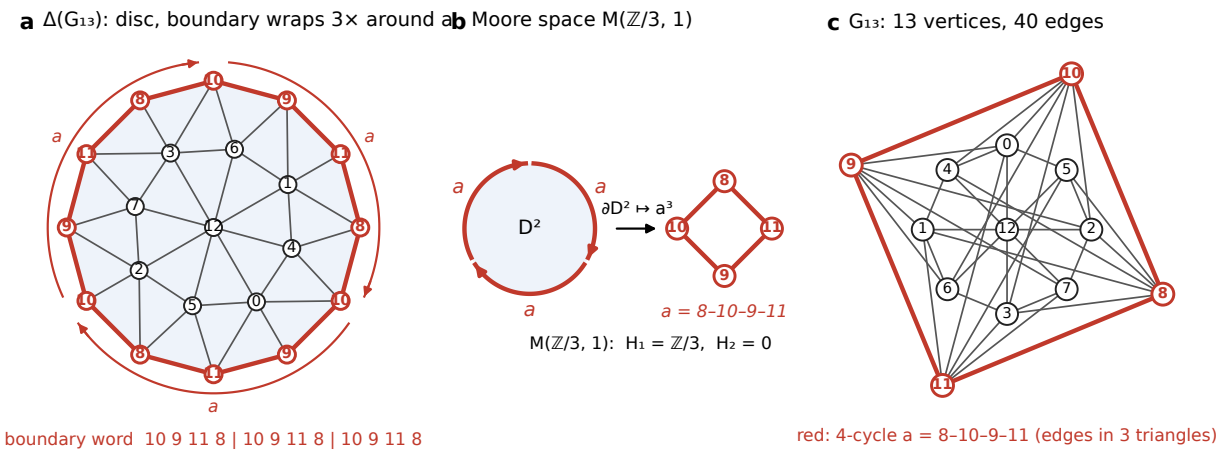



FIGURE 3. The graph $G_{13}$ on 13 vertices. (a) Its clique complex cut open along the four edges lying in three triangles (8–10, 10–9, 9–11, 11–8) is a disc with 28 triangles and 21 vertices whose boundary word is $(10\,9\,11\,8)^3$: the boundary winds three times around the 4-cycle in the same direction. (b) The resulting space is the Moore space $M(\mathbb{Z}/3,1)$. (c) A plane drawing of $G_{13}$ with vertex 12 (degree 8) at the centre, its neighbourhood $C_8$ around it and the 4-cycle in red.

## 6. EXAMPLES AND CONDITIONAL CLASSIFICATIONS

Throughout, vertices are labelled $0,\dots,n-1$ as in the `graph6` strings, except for Fig. 4(a), where the explicit map in the caption recovers the labels of [6].

### 6.1. Thirteen vertices: odd torsion

**Theorem 6.1.** *The following three graphs are $\mathbb{Z}_2$-acyclic and not strong I-contractible:*

$$G_{13} = \texttt{L?qa``\{wttym\~o}\ (40\ \textit{edges}),\qquad G_{13}+\{4,7\}\ (41),\qquad G_{13}+\{4,7\}+\{5,6\}\ (42).$$

*Their integral homology is $\mathbb{Z}/3$ in degree 1 and zero in all other positive degrees, and their clique complexes are homotopy equivalent to $M(\mathbb{Z}/3,1)$. None is I-contractible. The graph $G_{13}$ is stuck, has 28 triangles, and its $\mathrm{I}_4$-closure consists of these three graphs.*

*Under* H12, Hcasi *and* H13, *every $\mathbb{Z}_2$-acyclic graph on at most 12 vertices is in $\mathfrak{I}_S$, these are exactly the 13-vertex $\mathbb{Z}_2$-acyclic graphs outside $\mathfrak{I}_S$, and $G_{13}$ is the unique stuck graph of that order.*

*Proof.* Exact independent checks give the homology and strong-membership status of the three graphs. Complete-state expansion, followed by canonical labelling, gives the stated closure; their different edge counts show that they are pairwise non-isomorphic. The disc and boundary word described below identify $\Delta(G_{13})$ up to homotopy with $S^1 \cup_3 D^2$. The two admissible edge additions preserve homotopy type, so the same conclusion holds for the other two graphs. Nonzero integral homology excludes membership in $\mathfrak{I}$.

For completeness, assume the three stated hypotheses. Orders 1 and 2 are immediate; hypothesis H12 supplies the bound for the remaining orders through 12. Theorem 3.7 reduces the order-13 classification to the restricted enumeration in H13, whose only retained graph is $G_{13}$. Lemma 3.2(a) then places every order-13 exception in its three-element closure. □

The structure of $G_{13}$ is shown in Fig. 3. Its clique complex is pure 2-dimensional; exactly four edges lie in three triangles and they form the induced 4-cycle $C = 8$–10–9–11. Cutting along $C$ yields a disc whose boundary of length 12 is mapped onto $C$ with the word $(10\,9\,11\,8)^3$, that is, with the pattern $a^3$: $\Delta(G_{13}) \simeq M(\mathbb{Z}/3,1)$. The automorphism group has order 8, generated

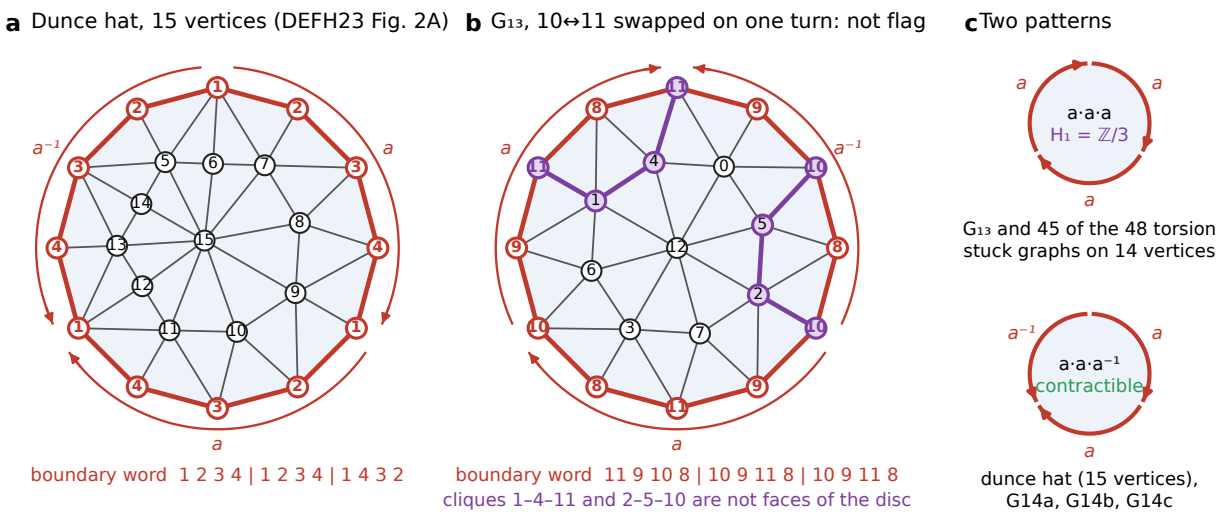



FIGURE 4. (a) The 15-vertex flag dunce hat $D_{15}$ of [6, Fig. 2A] unfolded: a disc with 23 vertices whose boundary of length 12 runs along the 4-cycle 1–2–3–4 with the word $1\,2\,3\,4 \mid 1\,2\,3\,4 \mid 1\,4\,3\,2$, i.e. the pattern $a\,a\,a^{-1}$ (labels $1,\dots,15$ of [6]; the `graph6` label $i$ corresponds to the label $(1,4,13,12,11,3,14,15,10,5,2,6,7,8,9)_i$, so this cycle is 0–10–5–1 in `graph6` labels). (b) Reversing one of the three passes of the boundary of $G_{13}$ (exchanging $10 \leftrightarrow 11$ in one block) gives a dunce hat, but its 1-skeleton contains the cliques 1–4–11 and 2–5–10 (purple) that are not faces of the disc, so the complex is not flag; the resulting graph is vertex-strong I-contractible. (c) The two identification patterns occurring among the stuck graphs of this paper.

by $(0\,1\,3\,2)(4\,6\,7\,5)(8\,10\,9\,11)$ and $(0\,1)(2\,3)(5\,6)(8\,10)(9\,11)$; the degree sequence is $5^4\,6^4\,7^4\,8$. The open neighbourhoods of the nine vertices off $C$ are induced cycles ($C_6$ for $0,1,2,3$; $C_5$ for $4,5,6,7$; $C_8$ for 12) and those of the four vertices of $C$ have seven vertices and eight edges (a cycle with a chord path), so no vertex is $\mathfrak{I}_S$-deletable; no common neighbourhood of an edge is $\mathbb{Z}_2$-acyclic either: since $G_{13}$ is $K_4$-free, each is an independent set, of size two or three according to the triangle incidence of the edge, so no edge is deletable—this is what "stuck" means.

*Remark* 6.2 (Torsion in small flag complexes). Adamaszek [19, Theorem 1.1 and Section 4] proves that no graph on at most ten vertices has torsion in $H_1$, and identifies four $H_1$-torsion graphs on eleven vertices and 363 vertex-irreducible ones on twelve vertices. Their torsion subgroup is $\mathbb{Z}/2$; five of the latter have an additional free summand. Every remaining twelve-vertex $H_1$-torsion graph $G$ has $G - v$ among the four eleven-vertex graphs. The Mayer–Vietoris sequence for $\Delta(G) = \Delta(G - v) \cup (v * \Delta(N(v)))$ shows that there is a short exact sequence

$$0 \longrightarrow Q \longrightarrow H_1(\Delta(G);\mathbb{Z}) \longrightarrow F \longrightarrow 0,$$

where $Q$ is a quotient of $\mathbb{Z}/2$ and $F$ is a subgroup of $\widetilde{H}_0(\Delta(N(v));\mathbb{Z})$. The group $F$ is free abelian, so the sequence splits and no odd torsion is introduced. Thus no flag complex on at most twelve vertices has odd torsion in $H_1$, without an acyclicity assumption. Our thirteen-vertex examples attain this bound. Under H12, Hcasi and H13, their classification here is exhaustive among $\mathbb{Z}_2$-acyclic flag complexes; no classification of all thirteen-vertex torsion graphs is claimed.

**Proposition 6.3.** *Under* H12, Hcasi *and* H13, *there is no graph on at most* 13 *vertices in* $\mathfrak{I} \setminus \mathfrak{I}_S$. *In particular, the dunce hat has no flag triangulation on at most* 13 *vertices.*

*Proof.* By Theorem 6.1, the only $\mathbb{Z}_2$-acyclic graphs on at most 13 vertices outside $\mathfrak{I}_S$ are the three torsion graphs. None is I-contractible, since its clique complex has nontrivial integral homology. If the dunce hat had a flag triangulation on at most 13 vertices, its graph would belong to $\mathfrak{I}$ by contractibility of the dunce hat, and hence to $\mathfrak{I}_S$ by the first assertion. Its clique complex would then be collapsible by [10, Theorem 1], contradicting the non-collapsibility of the dunce hat. □

It is instructive to see what goes wrong when one tries to turn $G_{13}$ into a dunce hat (Fig. 4(b)): reversing the orientation of one block of the boundary word changes the pattern to $a\,a\,a^{-1}$ and the resulting complex is contractible, but two triples of pairwise adjacent vertices are no longer faces, so the complex is not the clique complex of its 1-skeleton, and that 1-skeleton (the graph `L?qa``{wtlzM~o`, isomorphic to `LtDzQNKmpKXT`V`) contains the two copies $\{1,4,8,11\}$, $\{2,5,8,10\}$ of $K_4$ and is in $\mathfrak{I}_{VS}$.

### 6.2. **Fourteen vertices: examples of $\mathfrak{I} \setminus \mathfrak{I}_S$**

**Theorem 6.4.** *The* 51 *listed $K_4$-free graphs on* 14 *vertices are stuck. Exactly three have trivial reduced integer homology,*

$$G_{14}^a = \texttt{M?b@bFHNEk@~|k}\}\texttt{U?}, \qquad G_{14}^b = \texttt{M?B@thku?~Lylk}\}\texttt{[?},$$
$$G_{14}^c = \texttt{M?r@d`ifbiFfo}\}\texttt{~o?},$$

*with* 44, 45 *and* 45 *edges and* 31, 32 *and* 32 *triangles; each is I-contractible but not strong I-contractible. The remaining* 48 *have $H_1(\,\cdot\,;\mathbb{Z}) \cong \mathbb{Z}/3$. Under* H12, Hcasi, H13 *and* H14*, the list exhausts the stuck $K_4$-free graphs of order* 14*, the smallest graphs in $\mathfrak{I} \setminus \mathfrak{I}_S$ have order* 14*, and $G_{14}^a$ has the minimum number of edges among the $K_4$-free members of that order. Under these hypotheses there are exactly eight $K_4$-free graphs on* 14 *vertices in $\mathfrak{I} \setminus \mathfrak{I}_S$.*

*Proof.* Independent checks of the 51 listed graphs verify stuckness, pairwise non-isomorphism and integer homology. Under the stated hypotheses, Proposition 6.3 gives the lower bound on the order, and Theorem 3.7 together with H14 gives completeness in the $K_4$-free class. For $G_{14}^a$, the common neighbourhood of the non-adjacent vertices 0 and 1 is the induced path 8–12–5–10–13, which is in $\mathfrak{I}_S$, so $G_{14}^a + \{0,1\}$ is obtained from $G_{14}^a$ by a transformation (I4); Algorithm 2 finds a sequence of 18 strong deletions reducing $G_{14}^a + \{0,1\}$ to $K_1$ (Appendix C), so $G_{14}^a + \{0,1\} \in \mathfrak{I}_S \subseteq \mathfrak{I}$ and $G_{14}^a \in \mathfrak{I}$. Similarly $N_{G_{14}^b}(0,3) = 7$–13–9–12 and $N_{G_{14}^c}(2,3) = 6$–13–7–10–11 are induced paths and $G_{14}^b + \{0,3\}$, $G_{14}^c + \{2,3\}$ are in $\mathfrak{I}_S$ with 19 and 18 deletions. The three graphs are stuck, so not in $\mathfrak{I}_S$. Finally, a $K_4$-free graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ on 14 vertices with fewer than 44 edges would, by Lemma 3.2(b) (edge deletions preserve $K_4$-freeness), lie in the $\mathrm{I}_4$-closure of a stuck $K_4$-free graph with at most 43 edges; all such stuck graphs have $H_1 \cong \mathbb{Z}/3$, and (I4) additions preserve the homotopy type, a contradiction. The finite closure calculation below gives the count of eight; Lemma 3.2(b) supplies completeness under the stated hypotheses. □

Figures 5–7 show the three graphs. Their clique complexes are pure 2-dimensional and unfold into discs. For $G_{14}^a$ and $G_{14}^c$, the boundary is identified along a cycle with pattern $a\,a\,a^{-1}$; for $G_{14}^b$, the same pattern remains after contracting the pendant edge 9–13. The boundary maps therefore have degree $\pm 1$, giving a second proof of contractibility by Section 2. These complexes are not collapsible: every edge is contained in at least two triangles, so there is no free face with which to begin an elementary collapse (see Section 7.3). Table 3 records their cycle lengths 5, 4 and 6, their 44–45 edges, and their degrees between 5 and 8; the corresponding values for $D_{15}$ are 4, 46, and 4–10.

In each of the four examples, vertices outside the singular edges have cycle links. For $G_{14}^a$, $G_{14}^c$ and $D_{15}$, the singular set is precisely the identification cycle, whose vertex links have first Betti number two. For $G_{14}^b$, the singular set also contains 9–13: the link of vertex 9 has first Betti number two, and that of vertex 13 has first Betti number three. This accounts for the extra boundary excursions in Fig. 6.

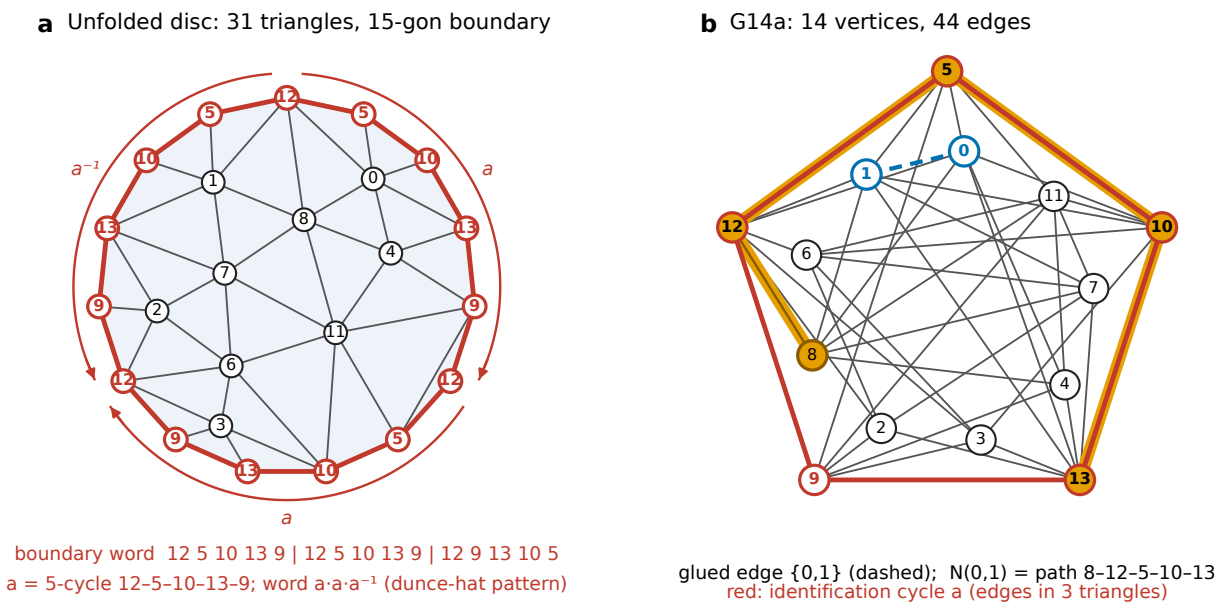



FIGURE 5. $G_{14}^{a}$ (44 edges, 31 triangles, trivial automorphism group). (a) Unfolded disc with 24 vertices; the boundary of length 15 is identified with the 5-cycle 5–10–13–9–12 with the dunce-hat pattern $a\,a\,a^{-1}$ (boundary word 12 5 10 13 9 | 12 5 10 13 9 | 12 9 13 10 5). (b) Plane drawing; the dashed edge $\{0,1\}$ is the (I4) addition certifying I-contractibility, and its common neighbourhood, the induced path 8–12–5–10–13, is highlighted.

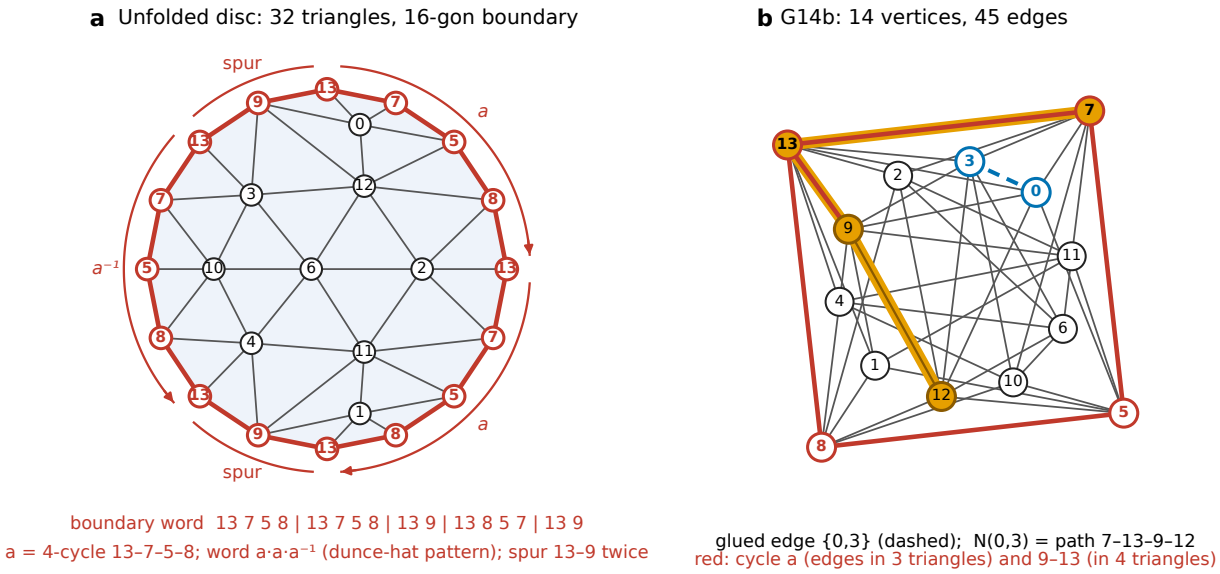



FIGURE 6. $G_{14}^{b}$ (45 edges, 32 triangles, $|\mathrm{Aut}| = 2$). The boundary of the unfolded disc (25 vertices, length 16) is identified with the 4-cycle 5–7–13–8 with pattern $a\,a\,a^{-1}$, and additionally runs back and forth along the pendant edge 13–9, which lies in four triangles. Each of its two back-and-forth excursions contributes zero to the degree of the boundary map. The (I4) edge is $\{0,3\}$ with $N(0,3) = 7$–13–9–12.

TABLE 3. The three 14-vertex examples and the dunce hat of [6]. $E_3$ = edges in exactly three triangles; disc = number of vertices of the unfolded disc; $k$ = length of the identification cycle. The additional pendant edge of $G_{14}^b$ belongs to four triangles and is not counted in $E_3$.

| | $n$ | $m$ | triangles | degrees | $\lvert E_3\rvert$ | $\lvert\mathrm{Aut}\rvert$ | disc | $k$ | pattern |
|---|---|---|---|---|---|---|---|---|---|
| $D_{15}$ | 15 | 46 | 32 | $4^3\,5^2\,6^7\,9^2\,10$ | 4 | 2 | 23 | 4 | $a\,a\,a^{-1}$ |
| $G_{14}^a$ | 14 | 44 | 31 | $5^3\,6^6\,7^3\,8^2$ | 5 | 1 | 24 | 5 | $a\,a\,a^{-1}$ |
| $G_{14}^b$ | 14 | 45 | 32 | $5^2\,6^5\,7^6\,8$ | 4 | 2 | 25 | 4 | $a\,a\,a^{-1}$ |
| $G_{14}^c$ | 14 | 45 | 32 | $6^9\,7^4\,8$ | 6 | 2 | 26 | 6 | $a\,a\,a^{-1}$ |

The 48 torsion examples are listed in Appendix B.1 (Table 4) and shown in Fig. 9. They have 43 (25 graphs), 44 (18), 45 (4) or 47 (1) edges. For 45 of them the unfolding is a disc whose boundary is identified with a cycle of length 4 (25 graphs), 5 (18) or 6 (2) with the pattern $a^3$, as for $G_{13}$; two unfold to an annulus and one (the graph with 47 edges, whose edges in three triangles form two disjoint 4-cycles) to an annulus together with a disc. These three cases are not covered by the single-disc argument of Section 2.

The $\mathrm{I}_4$-closures. Under H12, Hcasi, H13 and H14, Theorem 3.7(ii) shows that every $K_4$-free graph in $\mathfrak{I}\setminus\mathfrak{I}_S$ on 14 vertices lies in the $\mathrm{I}_4$-closure of one of the 51 stuck graphs (the closure itself may leave the $K_4$-free class). Computing the pruned closures of the 51 listed seeds gives 2,271 pairwise non-isomorphic $\mathbb{Z}_2$-acyclic graphs on 14 vertices outside $\mathfrak{I}_S$, of which 29 are contractible, hence in $\mathfrak{I}\setminus\mathfrak{I}_S$, and 2,242 have $H_1\cong\mathbb{Z}/3$ (file `data/acyclic_not_strong_14_closure.tsv` in the repository); 8 of the 29 are $K_4$-free, with 44–47 edges. Under the same computational hypotheses, these 8 graphs exhaust the $K_4$-free members of $\mathfrak{I}\setminus\mathfrak{I}_S$ on 14 vertices. These closures do not exhaust the $\mathbb{Z}_2$-acyclic graphs outside $\mathfrak{I}_S$ on 14 vertices. For example, adjoining a leaf to $G_{13}$ preserves its homotopy type and gives a graph outside $\mathfrak{I}_S$ with a deletable vertex. This graph cannot belong to any of the 51 closures: every enumerated seed has minimum degree at least four, whereas the new graph has a vertex of degree one. Lemma 3.2(a) does not apply at $n=14$, since the displayed 13-vertex example already gives $n_0\le 13$.

### 6.3. **Relation with the dunce hat of [6]**

The graph $D_{15}$ (Fig. 4(a)) is also stuck, with 46 edges, 32 triangles and the identification pattern $a\,a\,a^{-1}$ along the 4-cycle 0–10–5–1 (the 4-cycle 1–2–3–4 in the labels of [6, Fig. 2A]; the map is displayed in the caption of Fig. 4). It belongs to the same family as $G_{14}^a, G_{14}^b, G_{14}^c$: flag discs with identified boundary. The examples $G_{14}^a$ and $G_{14}^c$ realise the disc identification on one fewer vertex using cycles of lengths 5 and 6; $G_{14}^b$ uses a 4-cycle with an attached edge. Their non-isomorphic local structures show that the cycle length alone does not determine the number of vertices required. Adding the edge $\{9,12\}$ to $D_{15}$ (edge $\{5,7\}$ of [6, Fig. 2B]) produces a graph in $\mathfrak{I}_S$ with a 16-step deletion sequence (Appendix C), exactly as adding $\{0,1\}$ to $G_{14}^a$ does.

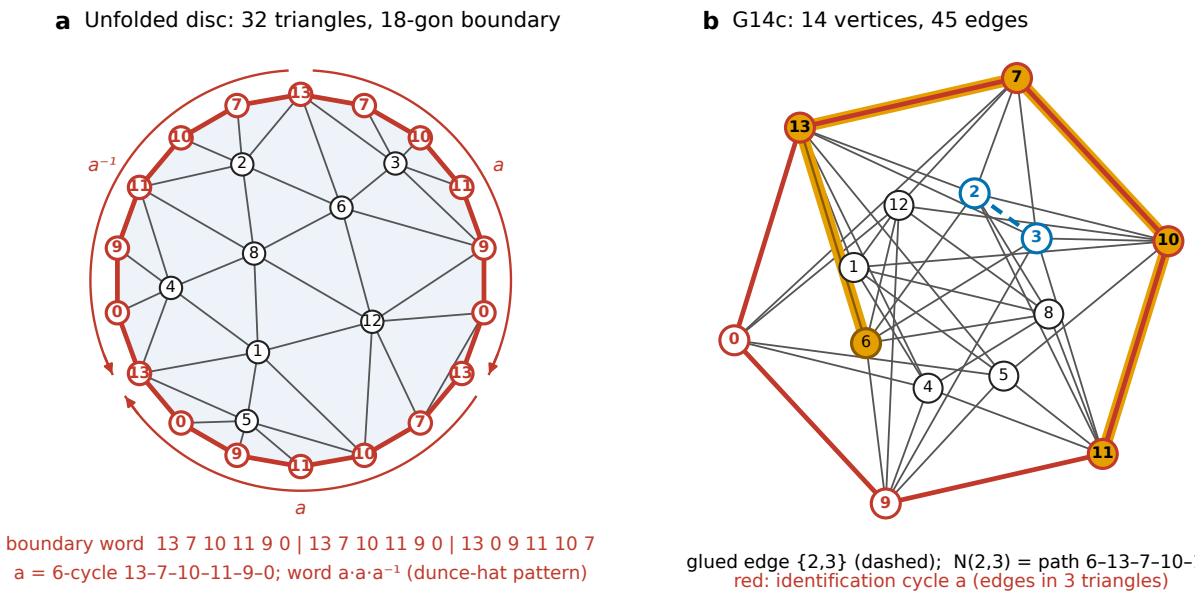



FIGURE 7. $G_{14}^{c}$ (45 edges, 32 triangles, $|\mathrm{Aut}| = 2$). The boundary of the unfolded disc (26 vertices, length 18) is identified with the 6-cycle 0–13–7–10–11–9 with pattern $a\,a\,a^{-1}$. The (I4) edge is $\{2,3\}$ with $N(2,3) = 6$–13–7–10–11.

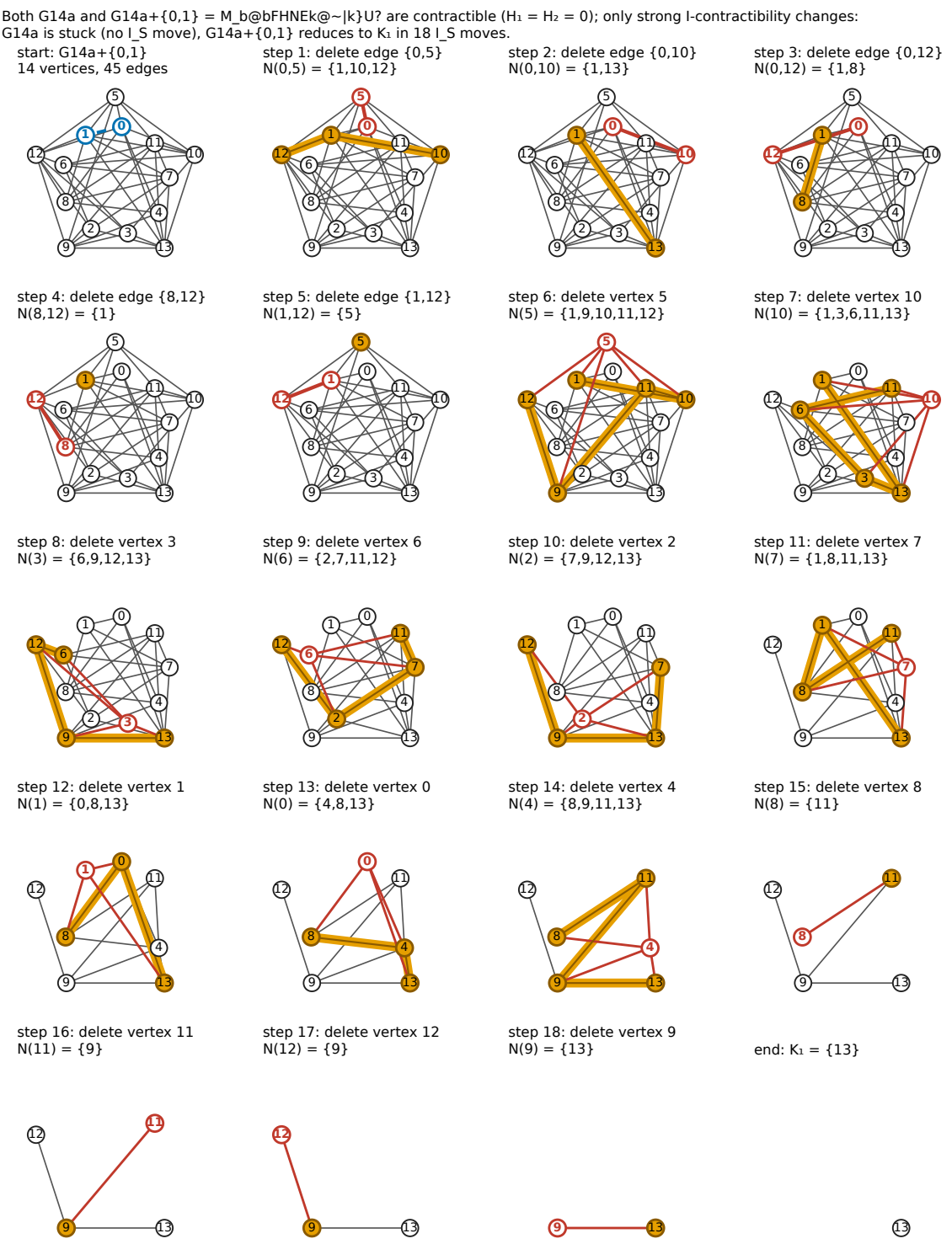



FIGURE 8. Certificate that $G_{14}^{a} + \{0,1\} \in \mathfrak{I}_S$: the 18 strong deletions found by Algorithm 2 (deleted element in red, its neighbourhood or common neighbourhood in amber). The sequence is listed in Appendix C and reproduced by `tools/strong_contractible.py --trace`.




FIGURE 9. The 51 stuck $\mathbb{Z}_2$-acyclic $K_4$-free graphs on 14 vertices returned by the recorded search (all such graphs under H14): the three contractible ones ($\mathfrak{I} \setminus \mathfrak{I}_S$) first, then the 48 with $H_1 \cong \mathbb{Z}/3$, ordered by number of edges. Red edges lie in a number of triangles other than two: for the discs they form the identification cycle $a$ (with a spur in $G_{14}^b$); for the two annuli ($*$), a 4-cycle and a disjoint edge in four triangles; for the annulus with a disc ($**$), two disjoint 4-cycles.

## 7. Consequences and open problems

### 7.1. Conjecture 15 of [6]

In a strong I-contractible graph the order of deletions matters [6, Fig. 2B]; Theorem 16 of [6] shows that for vertex-strong I-contractible graphs a bad choice of an $\mathfrak{I}_{VS}$-deletable vertex first occurs at 12 vertices. For edges, Conjecture 15 of [6] proposed that the first bad choice of an $\mathfrak{I}_S$-deletable edge occurs in the 15-vertex graph of their Fig. 2B (the dunce hat $D_{15}$ plus one edge).

**Theorem 7.1.** *Under* H12, Hcasi *and* H13, *the smallest graph $G$ (in terms of vertices) such that (1) $G \in \mathfrak{I}_S$ and (2) $G$ has an edge $\{v, w\}$ with $N(v, w) \in \mathfrak{I}_S$ and $G - \{v, w\} \notin \mathfrak{I}_S$ has* 14 *vertices. The graphs $G^a_{14} + \{0, 1\}$, $G^b_{14} + \{0, 3\}$ and $G^c_{14} + \{2, 3\}$ with graph6 strings*

```
M_b@bFHNEk@~|k}U?,  MCB@thku?~Lylk}[?,
M@r@d`ifbiFfo}~o?
```

*are such graphs independently of these hypotheses, and there are at least* 32 *pairwise non-isomorphic such graphs in the* $\mathrm{I}_4$*-closure of $G^a_{14}$ alone.*

*Proof.* $G^a_{14} + \{0, 1\} \in \mathfrak{I}_S$ with $N(0, 1) \in \mathfrak{I}_S$ and $(G^a_{14} + \{0, 1\}) - \{0, 1\} = G^a_{14} \notin \mathfrak{I}_S$ by Theorem 6.4; likewise for the other two. For minimality, if $G$ on $n \leq 13$ vertices satisfied (1) and (2), then $G - \{v, w\}$ would be $\mathbb{Z}_2$-acyclic (the deletion is a homotopy equivalence since $\Delta(N(v, w))$ is contractible) and not in $\mathfrak{I}_S$, so $n = 13$ and $G - \{v, w\}$ is one of the three graphs of Theorem 6.1, by Lemma 3.2(a) a graph in the $\mathrm{I}_4$-closure of $G_{13}$; but $G$ itself would then be in that closure, which contains no member of $\mathfrak{I}_S$. The count of 32 is the number of non-isomorphic graphs reached by one legal edge addition from a member outside $\mathfrak{I}_S$ in the $\mathrm{I}_4$-closure of $G^a_{14}$, and lying in $\mathfrak{I}_S$. It was recomputed by exhaustive expansion with full graph states and canonical labelling. Further additions from a graph in $\mathfrak{I}_S$ remain in $\mathfrak{I}_S$ by prepending the inverse edge deletion, so pruning those branches cannot miss any member outside $\mathfrak{I}_S$. The 32 boundary graphs therefore give 32 distinct examples; no claim is made that they exhaust all graphs with a bad edge deletion in the full closure. □

Under these hypotheses, "the order matters" for $\mathfrak{I}_S$-deletable edges first at 14 vertices, two more than for $\mathfrak{I}_{VS}$-deletable vertices.

### 7.2. The conjectures of [22]

Frías-Armenta [22] gave a 13-vertex counterexample to an axiom used by Ivashchenko and isolated three statements of [3] whose published proofs relied on it, restating them as conjectures: (Conjecture 3) every $G \in \mathfrak{I}$ on $n$ vertices can be transformed into the complete graph $K_n$ by additions of edges $uv$ with $N(u, v) \in \mathfrak{I}$; (Conjecture 4) if $G \in \mathfrak{I}$ and $G - v \in \mathfrak{I}$ then $N(v) \in \mathfrak{I}$; (Conjecture 5) if $G \in \mathfrak{I}$ then $N(v)$ and $G - v$ are I-homotopy equivalent. Our examples provide test cases where $\mathfrak{I}$ and $\mathfrak{I}_S$ differ, but they do not refute these conjectures.

**Proposition 7.2.** *Each of $G^a_{14}$, $G^b_{14}$, $G^c_{14}$ and $D_{15}$ can be transformed into a cone by* 11, 12, 12 *and* 8 *additions of edges $uv$ with $N(u, v) \in \mathfrak{I}_S$, respectively (sequences in Appendix C), and thereafter into $K_n$ by further such additions. For every vertex $v$ of these four graphs, $\widetilde{H}_*(G - v; \mathbb{Z}) \cong \widetilde{H}_*(N(v); \mathbb{Z})$, and no $G - v$ is contractible.*

*Proof.* The sequences in Appendix C verify the first assertion step by step. Once a universal vertex is present, every missing edge has a common neighbourhood containing that vertex as a cone point, so every remaining addition is legal. The homology isomorphism follows from the Mayer–Vietoris argument below. Each open neighbourhood of these four graphs has at most 10 vertices and is not $\mathbb{Z}_2$-acyclic (the neighbourhood checks are included in the certificates). Hence $G - v$ is not $\mathbb{Z}_2$-acyclic and cannot be contractible. □

The second statement is not a coincidence: if $\Delta(G)$ is contractible then $\Delta(G) = \Delta(G - v) \cup \mathrm{cone}(\Delta(N(v)))$ with intersection $\Delta(N(v))$, and the Mayer–Vietoris sequence gives $\widetilde{H}_k(G - v) \cong \widetilde{H}_k(N(v))$ for all $k$. Hence Conjectures 4 and 5 cannot be refuted by homological invariants on graphs in $\mathfrak{I}$; a counterexample would have to be detected by the fundamental group or by simple homotopy type. (By [5, Theorem 1.2], $\mathfrak{I}$-homotopy equivalence of graphs is simple homotopy equivalence of clique complexes, so Conjecture 5 asks whether $\Delta(G - v)$ and $\Delta(N(v))$ are simple homotopy equivalent whenever $\Delta(G)$ is contractible.)

### 7.3. **Open problems**

**Question 7.3.** Do the 29 graphs in $\mathfrak{I} \setminus \mathfrak{I}_S$ obtained from the closures in Section 6.2 exhaust this class on 14 vertices? Under the computational hypotheses above, this is equivalent to asking whether every stuck graph in $\mathfrak{I} \setminus \mathfrak{I}_S$ on 14 vertices is $K_4$-free.

Among the 29 known graphs, 21 already contain a $K_4$; thus the corresponding question for all graphs in $\mathfrak{I} \setminus \mathfrak{I}_S$, rather than its stuck members, has a negative answer. The full class of 2-connected graphs on 14 vertices with $4 \leq \delta \leq \Delta \leq 12$ has about $10^{16}$ members, two to three orders of magnitude beyond the present computation. Two routes seem feasible: enumerating only graphs with a bounded number of copies of $K_4$, or developing further structural restrictions on stuck graphs containing a $K_4$. Adding one vertex to known smaller graphs is useful for discovering examples, but does not by itself give an exhaustive search.

**Question 7.4.** Among stuck $K_4$-free graphs of a fixed order, consider those with the minimum number of edges, separately in the contractible case and in the non-contractible case. Must their clique complexes be quotients of a disc with boundary attached to a unicyclic graph, with degree $\pm 1$ or $\pm 3$ on its unique cycle?

No such statement holds for all stuck graphs: among the 51 examples on 14 vertices, two unfold to annuli and one has a disconnected unfolding. Here the cutting set consists of all edges whose triangle incidence differs from two, as explained in Section 2.

**Proposition 7.5.** *The clique complexes of $G_{14}^a$, $G_{14}^b$ and $G_{14}^c$ are not collapsible.*

*Proof.* Each is a pure 2-dimensional complex and every edge belongs to at least two triangles. Consequently there is no free edge–triangle pair; purity also excludes a free vertex–edge pair. More generally no vertex can be contained in a unique maximal triangle, since each of its incident edges lies in at least two triangles. Thus no elementary collapse can start. These complexes have more than one vertex, so they are not collapsible. □

The conjecture that a flag complex is collapsible if and only if its 1-skeleton belongs to $\mathfrak{I}_S$ is Conjecture 1 of [10], restated as Conjecture 27 of [6]. The forward implication remains a separate question; the implication $\mathfrak{I}_S \Rightarrow$ collapsible is [10, Theorem 1]. The three graphs above therefore do not refute that conjecture.

**Question 7.6.** Which first-homology torsion groups occur in flag complexes on 13 and 14 vertices? In particular, which 13-vertex examples with odd torsion fail to be $\mathbb{Z}_2$-acyclic?

The minimum order for odd torsion in $H_1$ is 13, even without the acyclicity assumption; see the discussion of Adamaszek's classification in Section 6.1.

**Question 7.7.** What is the smallest order of a graph in $\mathfrak{I}_{VS}$ that is not 2-dismantlable?

Such graphs exist: the graph $Q_4$ of [7, Proposition 8] has 24 vertices and belongs to 3-Dis but not 2-Dis. Thus the minimum is at most 24; if the reported rows of orders 12 and 13 are complete, it is at least 14 (Observation 5.1(ii)).

Finally, Question 28 of [6] asked whether every graph with contractible clique complex is I-contractible; this is now the theorem of Dochtermann and Matsushita [5]. Our results show that the analogous question with "$\mathbb{Z}_2$-acyclic" in place of "contractible" has a negative answer already at 13 vertices, and, under the stated computational hypotheses, that the gap between $\mathfrak{I}$ and $\mathfrak{I}_S$ opens at 14.

## 8. Data and code availability

The public repository (Section 1) contains the C implementation (`src/`), run-specific compiled sources and archived scripts (`yuca/`), aggregate counters and slice registries, the graph lists (`data/`), independent Python implementations (`tools/`), regenerated certificates, figures and manuscript sources, and the version-2 classifier and campaign pipeline prepared for the recomputations (`v2/`). Commit `4a059b5` is the state examined in the first round of the external audit. The per-slice output archive is retained on YUCA; the public repository contains summaries and registries rather than every raw output file. In commit `4a059b5` the p011 worker in `yuca/as_run/` is misidentified; since commit `892e41b` the repository holds the identified worker under that name and keeps the misattributed file unchanged under another name (Appendix D); a copy of the identified worker also accompanies the manuscript sources (`appendix_data/`). The release that accompanies publication will be archived with a persistent identifier.

The repository records the independent checks: the Python verification written from scratch (`verification/indep.py`) and its report, the Wolfram Language code and outputs (`verification/wolfram_codigo_y_salidas.json`), and the comparison of the manuscript with the data (`paper/NUMERIC_CHECK.md`). The script `verify.sh` checks listed finite certificates and, when `gcc` and `geng` are available, the C results and table rows through order 9; `--full-table` requests rows 10 and 11 as well. Missing dependencies cause levels to be skipped, so exit status 0 must be interpreted together with the log. With both dependencies present it passes all 39 checks (16 October 2026). It does not certify the original large enumerations or every structural claim. Code is released under the MIT licence and data under CC BY 4.0.

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## Appendix A. Proofs of the reduction lemmas

We use two facts. (F1) Put $L = \Delta(N(v))$. Then $\Delta(G) = \Delta(G - v) \cup (v * L)$ and the intersection is $L$. If $L$ is non-empty and contractible, both $L$ and $v * L$ are contractible; gluing along these simplicial subcomplexes shows that $\Delta(G - v) \hookrightarrow \Delta(G)$ is a homotopy equivalence. For an edge $uv$, put $L = \Delta(N(u, v))$. Then

$$\Delta(G) = \Delta(G - uv) \cup ((uv) * L), \qquad \Delta(G - uv) \cap ((uv) * L) = (\partial(uv)) * L.$$

If $L$ is non-empty and contractible, both the join $(uv) * L$ and its attaching subcomplex $(\partial(uv)) * L$ are contractible, and the same gluing argument applies. Thus deleting an $\mathfrak{I}_S$-deletable element preserves homotopy type and acyclicity. It also preserves membership in $\mathfrak{I}$, directly by (I1) or

(I3), since $\mathfrak{I}_S \subseteq \mathfrak{I}$. The reverse additions preserve these properties as well. (F2) If $H \in \mathfrak{I}_S$ and $v$ (respectively $uv$) has neighbourhood (respectively common neighbourhood) in $\mathfrak{I}_S$ in $G$, with $G - v = H$ (respectively $G - uv = H$), then $G \in \mathfrak{I}_S$: prepend this deletion to a reduction of $H$. This implication does not assert that every legal deletion from a member of $\mathfrak{I}_S$ remains in $\mathfrak{I}_S$.

*Proof of Lemma 3.2.* (a) Let $G$ be $\mathbb{Z}_2$-acyclic, not in $\mathfrak{I}_S$, on $n_0$ vertices. If $G$ has an $\mathfrak{I}_S$-deletable vertex $v$, then $G - v$ is $\mathbb{Z}_2$-acyclic (F1) and not in $\mathfrak{I}_S$ (F2), with $n_0 - 1$ vertices, contradicting the minimality of $n_0$. If $G$ has an $\mathfrak{I}_S$-deletable edge $uv$, then $G - uv$ is $\mathbb{Z}_2$-acyclic and not in $\mathfrak{I}_S$ with the same number of vertices and one edge fewer, and $G = (G - uv) + uv$ is an (I4) addition. Induction on the number of edges shows that $G$ lies in the $\mathrm{I}_4$-closure of a graph with no $\mathfrak{I}_S$-deletable element, which is stuck; if $G$ has the minimum number of edges, it has no $\mathfrak{I}_S$-deletable edge and is itself stuck.

(b) Let $G \in \mathfrak{I} \setminus \mathfrak{I}_S$ on $n_1$ vertices. If $v$ is $\mathfrak{I}_S$-deletable then $G - v \in \mathfrak{I}$ (transformation (I1), since $\mathfrak{I}_S \subseteq \mathfrak{I}$) and $G - v \notin \mathfrak{I}_S$ (F2), contradicting the minimality of $n_1$. If $uv$ is $\mathfrak{I}_S$-deletable then $G - uv \in \mathfrak{I} \setminus \mathfrak{I}_S$ by (I3) and (F2), has one edge fewer and is $K_4$-free if $G$ is. Induction on the number of edges concludes as in (a); the graph reached is $\mathbb{Z}_2$-acyclic because it is in $\mathfrak{I}$. □

*Proof of Lemma 3.3.* Let $G = A \cup_u B$ be stuck. Since $\Delta(G) = \Delta(A) \cup \Delta(B)$ with $\Delta(A) \cap \Delta(B) = \{u\}$ contractible, the Mayer–Vietoris sequence gives $\widetilde{H}_*(G;\mathbb{Z}_2) \cong \widetilde{H}_*(A;\mathbb{Z}_2) \oplus \widetilde{H}_*(B;\mathbb{Z}_2)$, so $A$ and $B$ are $\mathbb{Z}_2$-acyclic. Because $u$ is a cut vertex, for every $w \in V(A) \setminus \{u\}$ we have $N_G(w) = N_A(w)$, and for every edge $e$ of $A$ we have $N_G(e) = N_A(e)$ (a common neighbour of the ends of $e$ lies in $A$). Since $G$ is stuck, no vertex $w \neq u$ and no edge of $A$ is $\mathfrak{I}_S$-deletable in $A$: $A$ is almost stuck at $u$, and likewise $B$. If $A$ had only two vertices, $A = uw$ and $N_G(w) = \{u\} \cong K_1 \in \mathfrak{I}_S$, contradicting that $G$ is stuck; so $|A|, |B| \geq 3$. If no almost stuck graph has at most $m$ vertices then $|A|, |B| \geq m + 1$ and $|G| = |A| + |B| - 1 \geq 2m + 1$, which proves the last statement. □

*Proof of Lemma 3.4.* Let $uv$ be an edge in no triangle. Since $G$ is 2-connected, $uv$ lies in a cycle $C$. The 1-chain $C$ (over $\mathbb{Z}_2$) is a cycle; the boundary of any 2-chain has coefficient 0 on $uv$ because no triangle contains $uv$, so $C$ is not a boundary and $\widetilde{H}_1(G;\mathbb{Z}_2) \neq 0$, a contradiction. Hence every edge lies in a triangle and every $N(u,v)$ is non-empty; since $G$ is stuck, $N(u,v) \notin \mathfrak{I}_S$, so $|N(u,v)| \geq 2$. Now fix $v$ and let $H = N(v)$, a graph with $|H| \geq 1$ and $H \notin \mathfrak{I}_S$, in which every vertex $u$ satisfies $N_H(u) = N(u,v) \neq \emptyset$ with $|N_H(u)| \geq 2$. A graph with at most three vertices in which every vertex has at least two neighbours is $K_3 \in \mathfrak{I}_S$; hence $|H| \geq 4$, i.e. $\delta(G) \geq 4$. Finally, if $\deg v = n - 1$ then $G$ is a cone with apex $v$, which is 0-dismantlable and hence in $\mathfrak{I}_S$, a contradiction; so $\Delta(G) \leq n - 2$. □

## Appendix B. Graph lists

### B.1. The 51 stuck $K_4$-free graphs on 14 vertices

Table 4: The 51 stuck $\mathbb{Z}_2$-acyclic $K_4$-free graphs on 14 vertices. $m$ = edges, $t$ = triangles; $H_1 = H_1(\Delta(G);\mathbb{Z})$; "$\mathfrak{I}$" gives the $\mathrm{I}_4$ edge certifying I-contractibility; $k$ = length of the cycle carrying the boundary identification and "pattern" its orientation word ($a^3$ or $a\,a\,a^{-1}$), when the unfolding is a disc.

| # | graph6 | $m$ | $t$ | $H_1$ | $\mathfrak{I}$ | \|Aut\| | $k$ / pattern |
|---|---|---|---|---|---|---|---|
| 1 | `M?b@bFHNEk@~\|k}U?` | 44 | 31 | 0 | ({0,1}) | 1 | 5 / $a\,a\,a^{-1}$ |
| 2 | `M?B@thku?~Lylk}[?` | 45 | 32 | 0 | ({0,3}) | 2 | 4 / $a\,a\,a^{-1}$ (+spur) |
| 3 | `M?r@d`ifbiFfo}~o?` | 45 | 32 | 0 | ({2,3}) | 2 | 6 / $a\,a\,a^{-1}$ |
| 4 | `M?B@cXamAu\r{]V{?` | 43 | 30 | $\mathbb{Z}/3$ | no | 2 | 4 / $a^3$ |
| 5 | `M?B@cXemAuB}ze{]?` | 43 | 30 | $\mathbb{Z}/3$ | no | 2 | 4 / $a^3$ |
| 6 | `M?BDA_wqpmJvj]\|w?` | 43 | 30 | $\mathbb{Z}/3$ | no | 2 | 4 / $a^3$ |
| 7 | `M?`@E_wDf[UnU}z{?` | 43 | 30 | $\mathbb{Z}/3$ | no | 6 | 4 / $a^3$ |
| 8 | `M?`adQWT\LK^du^w?` | 43 | 30 | $\mathbb{Z}/3$ | no | 2 | 4 / $a^3$ |

| # | graph6 | $m$ | $t$ | $H_1$ | $\mathfrak{I}$ | \|Aut\| | $k$ / pattern |
|---|---|---|---|---|---|---|---|
| 9 | M?`adQWtVDUr[]^w? | 43 | 30 | $\mathbb{Z}/3$ | no | 2 | 4 / $a^3$ |
| 10 | M?bB@_WmbRJZ\|M~w? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 11 | M?bB@_WsbX^Jlm~w? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 12 | M?bB@_WsequuZZ~w? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 13 | M?bB@_Wu`ZJjlm~w? | 43 | 30 | $\mathbb{Z}/3$ | no | 2 | 4 / $a^3$ |
| 14 | M?bB@_WubXNJlm~w? | 43 | 30 | $\mathbb{Z}/3$ | no | 2 | 4 / $a^3$ |
| 15 | M?bB@aWYTkZZ^M\|w? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 16 | M?bB@aWZSmXZ\|wNm? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 17 | M?bB@aWZTkFf\|wuu? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 18 | M?bB@aWdbYFv\|wuu? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 19 | M?bB@aWdbZEv\|wuu? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 20 | M?bB@aWqplQzZ]\|w? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 21 | M?bB@aWyTkYZ^M\|w? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 22 | M?bB@aXdbY[{Nmuu? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 23 | M?bB@aXlbYFfxwuu? | 43 | 30 | $\mathbb{Z}/3$ | no | 2 | 4 / $a^3$ |
| 24 | M?bB@aXyTkYZxw^M? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 25 | M?otU`SmVEKm]M^o? | 43 | 30 | $\mathbb{Z}/3$ | no | 4 | 4 / $a^3$ |
| 26 | M?qa``{rfEEymitY? | 43 | 30 | $\mathbb{Z}/3$ | no | 4 | 4 / $a^3$ |
| 27 | M?qa``{ufEUimitY? | 43 | 30 | $\mathbb{Z}/3$ | no | 4 | 4 / $a^3$ |
| 28 | M?r@daiRbrTYd{tb? | 43 | 30 | $\mathbb{Z}/3$ | no | 1 | 4 / $a^3$ |
| 29 | M?`Db_kVcyl]hyvw? | 44 | 31 | $\mathbb{Z}/3$ | no | 2 | 5 / $a^3$ |
| 30 | M?`Db_khi{L\nUvw? | 44 | 31 | $\mathbb{Z}/3$ | no | 1 | 5 / $a^3$ |
| 31 | M?`adL[mBkvLY{d{? | 44 | 31 | $\mathbb{Z}/3$ | no | 2 | 5 / $a^3$ |
| 32 | M?`cu_whRRFZ~Wtm? | 44 | 31 | $\mathbb{Z}/3$ | no | 1 | 5 / $a^3$ |
| 33 | M?`cv@UNDNZMtsY{? | 44 | 31 | $\mathbb{Z}/3$ | no | 1 | 5 / $a^3$ |
| 34 | M?b@bCw`usLvvw{]? | 44 | 31 | $\mathbb{Z}/3$ | no | 1 | 5 / $a^3$ |
| 35 | M?qa``[NNEYlme~o? | 44 | 31 | $\mathbb{Z}/3$ | no | 2 | 5 / $a^3$ |
| 36 | M?qa``[iUEylNm~o? | 44 | 31 | $\mathbb{Z}/3$ | no | 1 | 5 / $a^3$ |
| 37 | M?qa``[icjk^Nk~o? | 44 | 31 | $\mathbb{Z}/3$ | no | 2 | 5 / $a^3$ |
| 38 | M?qa``[iej[ZNk~o? | 44 | 31 | $\mathbb{Z}/3$ | no | 2 | 5 / $a^3$ |
| 39 | M?qa``kLeE^wt]i}? | 44 | 31 | $\mathbb{Z}/3$ | no | 2 | 5 / $a^3$ |
| 40 | M?qa``kMmMIniu~o? | 44 | 31 | $\mathbb{Z}/3$ | no | 2 | 5 / $a^3$ |
| 41 | M?qa``kifEfUtZ~o? | 44 | 31 | $\mathbb{Z}/3$ | no | 1 | 5 / $a^3$ |
| 42 | M?qa`awkmiNdwu~o? | 44 | 31 | $\mathbb{Z}/3$ | no | 2 | 5 / $a^3$ |
| 43 | M?qa`awwRxYj{k~o? | 44 | 31 | $\mathbb{Z}/3$ | no | 2 | 5 / $a^3$ |
| 44 | M?qa`g{guLzwmkU]? | 44 | 31 | $\mathbb{Z}/3$ | no | 2 | 5 / $a^3$ |
| 45 | M?qa`jSwubncvomd? | 44 | 31 | $\mathbb{Z}/3$ | no | 2 | 5 / $a^3$ |
| 46 | M?qaejaLe[ngmh{[? | 44 | 31 | $\mathbb{Z}/3$ | no | 2 | 5 / $a^3$ |
| 47 | M?`FSzcVBmVFs{J{? | 45 | 32 | $\mathbb{Z}/3$ | no | 4 | 6 / $a^3$ |
| 48 | M?qa``{ws}Lxmkt[? | 45 | 32 | $\mathbb{Z}/3$ | no | 8 | annulus |
| 49 | M?qa``{wvkVUt[}Q? | 45 | 32 | $\mathbb{Z}/3$ | no | 8 | annulus |
| 50 | MCOfczgVB]UeM{p{? | 45 | 32 | $\mathbb{Z}/3$ | no | 4 | 6 / $a^3$ |
| 51 | MCp`faiFf{W}][js? | 47 | 34 | $\mathbb{Z}/3$ | no | 2 | not a disc |

## B.2. **The twenty graphs of order** 12 **in 2-Dis\1-Dis**

Table 5 lists the 20 connected graphs on 12 vertices that are 2-dismantlable but not 1-dismantlable, as returned by a separate run in mode `2dis_no_1dis` over all 164,059,830,476 connected graphs of order 12 (4,000 slices, none missing; the total agrees with Table 2). Each listed graph is verified individually: its 2-dismantling sequence is in the repository, and the exact decision of the C program and the independent Python implementation both show that it is not 1-dismantlable. That no other graph of order 12 has this property is subject to the same premises as row 12 of Table 2 (H12 and Appendix D).

TABLE 5. The 20 connected graphs on 12 vertices in 2-Dis $\setminus$ 1-Dis: $m$ edges, $t$ triangles, the number of $K_4$ subgraphs, clique number $\omega$, minimum and maximum degree $\delta, \Delta$, and the order of the automorphism group.

| # | graph6 | $m$ | $t$ | $K_4$ | $\omega$ | $\delta, \Delta$ | \|Aut\| |
|---|---|---|---|---|---|---|---|
| 1 | KEhutxmv\|~^{ | 46 | 69 | 34 | 4 | 6,9 | 4 |
| 2 | KEjer^{z}~^{ | 47 | 73 | 37 | 4 | 6,9 | 2 |
| 3 | KQyurzU~t~^{ | 47 | 72 | 37 | 5 | 7,9 | 4 |
| 4 | KQyurzU~u~^{ | 47 | 73 | 39 | 5 | 6,9 | 2 |
| 5 | KEhvtzm\|\|~^{ | 48 | 78 | 45 | 5 | 6,9 | 2 |
| 6 | KEjeu~m^~z^r | 48 | 77 | 42 | 5 | 6,9 | 1 |
| 7 | KEnfbzm\|\|~^{ | 48 | 75 | 38 | 4 | 7,9 | 4 |
| 8 | KEnfbzm~]~^{ | 48 | 75 | 38 | 4 | 7,9 | 2 |
| 9 | KEnfbzm~^v^{ | 48 | 75 | 38 | 4 | 7,9 | 2 |
| 10 | KQyurzuv\|~^{ | 48 | 77 | 44 | 5 | 7,9 | 4 |
| 11 | KQyurzuv~}^{ | 48 | 76 | 42 | 5 | 7,9 | 2 |
| 12 | KEht~jzz~}^f | 49 | 82 | 50 | 5 | 7,9 | 2 |
| 13 | KEjflzz^~}^f | 49 | 81 | 47 | 5 | 7,9 | 2 |
| 14 | KEnb~jl~t~^{ | 49 | 81 | 48 | 5 | 7,9 | 2 |
| 15 | KEnb~jl~v\|^{ | 49 | 80 | 45 | 5 | 8,9 | 2 |
| 16 | KEnff^{^~v^{ | 49 | 80 | 44 | 5 | 7,9 | 1 |
| 17 | KEnfnZ{~r~^{ | 49 | 79 | 42 | 5 | 8,9 | 4 |
| 18 | KUZurzm~vv^{ | 49 | 81 | 48 | 5 | 8,9 | 4 |
| 19 | KEnb~rvz~}^f | 50 | 86 | 54 | 5 | 8,9 | 2 |
| 20 | KEzftzv^~}^f | 50 | 85 | 51 | 5 | 8,9 | 2 |

## APPENDIX C. CERTIFICATES

### C.1. **Strong deletion sequences**

The following sequences, produced by `tools/strong_contractible.py --trace` (an exact search independent of the C program), reduce $G_{14}^a + \{0,1\}$, $G_{14}^b + \{0,3\}$, $G_{14}^c + \{2,3\}$ and $D_{15} + \{9,12\}$ to $K_1$. Each line deletes a vertex $v$ with $N(v) \in \mathfrak{I}_S$ or an edge $uv$ with $N(u,v) \in \mathfrak{I}_S$ in the current graph; the reader may check each step with any of the tools in the repository.

```
# Deletion sequences certifying strong I-contractibility (J_S) of G + e
# for the three minimal examples (tools/strong_contractible.py, exact search).
# Vertex labels are those of the graph6 string. Each step deletes a vertex v
# with N(v) in J_S, or an edge uv with N(u) cap N(v) in J_S, in the current graph.

M?b@bFHNEk@~|k}U? +{0,1}  n=14 m=45  in J_S
   1. delete edge 0-5
   2. delete edge 0-10
   3. delete edge 0-12
   4. delete edge 8-12
   5. delete edge 1-12
   6. delete vertex 5
   7. delete vertex 10
   8. delete vertex 3
   9. delete vertex 6
  10. delete vertex 2
  11. delete vertex 7
  12. delete vertex 1
  13. delete vertex 0
  14. delete vertex 4
  15. delete vertex 8
  16. delete vertex 11
  17. delete vertex 12
  18. delete vertex 9
  -> K_1 after 18 deletions
```

```
M?B@thku?~Lylk}[? +{0,3}  n=14 m=46  in J_S
   1. delete edge 0-9
   2. delete edge 0-13
   3. delete edge 9-12
   4. delete edge 3-9
   5. delete edge 3-13
   6. delete edge 7-13
   7. delete vertex 2
   8. delete vertex 6
   9. delete vertex 3
  10. delete vertex 0
  11. delete vertex 4
  12. delete vertex 7
  13. delete vertex 5
  14. delete vertex 1
  15. delete vertex 10
  16. delete vertex 11
  17. delete vertex 9
  18. delete vertex 12
  19. delete vertex 8
  -> K_1 after 19 deletions

M?r@d`ifbiFfo}~o? +{2,3}  n=14 m=46  in J_S
   1. delete edge 2-7
   2. delete edge 2-10
   3. delete edge 2-13
   4. delete edge 6-13
   5. delete edge 3-13
   6. delete vertex 7
   7. delete vertex 10
   8. delete vertex 1
   9. delete vertex 4
  10. delete vertex 0
  11. delete vertex 8
  12. delete vertex 2
  13. delete vertex 3
  14. delete vertex 6
  15. delete vertex 11
  16. delete vertex 12
  17. delete vertex 9
  18. delete vertex 5
  -> K_1 after 18 deletions

# Dunce hat (Fig. 2A of DEFH 2023) plus the edge of Fig. 2B:
N|thHsL`sHoI`RPOwLG +{9,12}  n=15 m=47  in J_S
   1. delete vertex 11
   2. delete edge 0-9
   3. delete edge 0-12
   4. delete vertex 10
   5. delete vertex 8
   6. delete vertex 4
   7. delete vertex 0
   8. delete vertex 1
   9. delete vertex 2
  10. delete vertex 3
  11. delete vertex 5
  12. delete vertex 6
  13. delete vertex 7
  14. delete vertex 9
  15. delete vertex 12
  16. delete vertex 13
  -> K_1 after 16 deletions
```

Figures 10 and 11 show the sequences for $G^b_{14} + \{0,3\}$ and $G^c_{14} + \{2,3\}$ step by step, as Fig. 8 does for $G^a_{14} + \{0,1\}$.

### C.2. **Routes to a cone (Proposition 7.2)**

Adding the following edges in order, each with common neighbourhood in $\mathfrak{I}_S$ at the moment of addition, turns the graph into a cone with the indicated apex (verified by `tools/strong_contractible.py`):

- $G^a_{14}$: $\{0,1\}$, $\{8,13\}$, $\{0,7\}$, $\{1,4\}$, $\{4,7\}$, $\{7,9\}$, $\{6,9\}$, $\{6,13\}$, $\{11,13\}$, $\{5,13\}$, $\{12,13\}$; apex 13.
- $G^b_{14}$: $\{0,3\}$, $\{3,5\}$, $\{10,12\}$, $\{0,10\}$, $\{0,6\}$, $\{6,8\}$, $\{1,10\}$, $\{1,4\}$, $\{10,11\}$, $\{2,10\}$, $\{10,13\}$, $\{9,10\}$; apex 10.
- $G^c_{14}$: $\{0,11\}$, $\{2,3\}$, $\{2,9\}$, $\{6,11\}$, $\{8,9\}$, $\{0,8\}$, $\{3,8\}$, $\{0,1\}$, $\{0,10\}$, $\{0,2\}$, $\{0,3\}$, $\{0,6\}$; apex 0.
- $D_{15}$: $\{2,4\}$, $\{3,5\}$, $\{2,8\}$, $\{9,12\}$, $\{5,7\}$, $\{1,7\}$, $\{7,10\}$, $\{0,7\}$; apex 7.

### C.3. **Integer homology**

`certificates/integer_homology.txt` lists, for the three graphs of Theorem 6.1, the 51 stuck graphs and $D_{15}$, the number of simplices in each dimension, the $\mathbb{Z}_2$-Betti numbers and the integer homology groups computed by Smith normal form with `tools/clique_homology.py`.

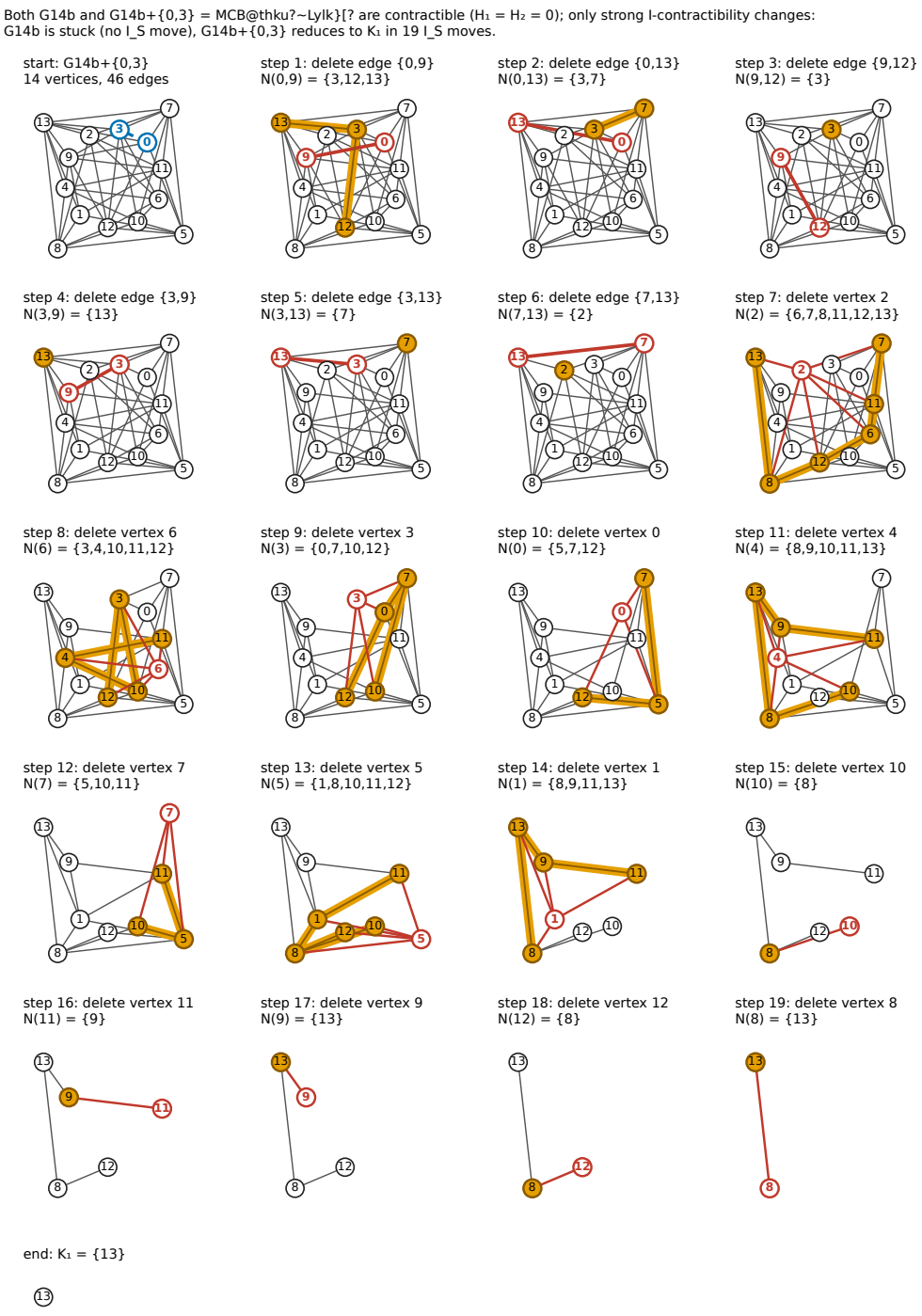



FIGURE 10. Certificate that $G_{14}^{b} + \{0,3\} \in \mathfrak{I}_S$ (19 strong deletions).

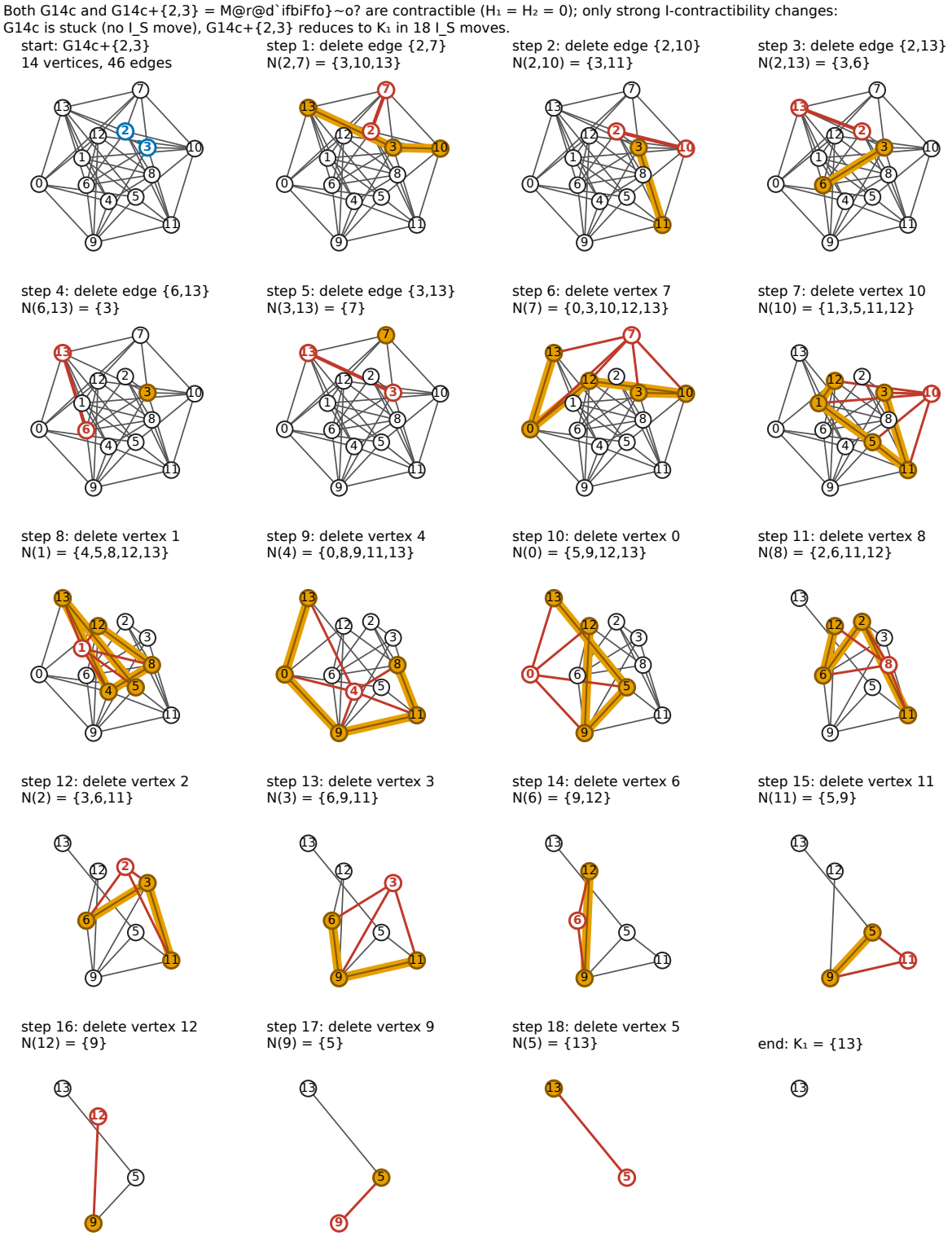



FIGURE 11. Certificate that $G_{14}^c + \{2,3\} \in \mathfrak{I}_S$ (18 strong deletions).

## Appendix D. Details of the cluster computation

The CPU partition of YUCA comprises 24 nodes with two 32-core AMD EPYC 9354 processors per node, 1 TB of RAM and no hyper-threading. The archived system records describe Rocky Linux 9.6, SLURM 24.11 and GCC 11. The order-13 run used `nauty` 2.7r4; the valid $K_4$-free order-14 run used the locally compiled version 2.8. The two projects each had a 500-core quality-of-service limit. Algorithm 4 summarises the scheduling design; the qualifications below describe the actual concurrent workers.

Coverage and counters. The seven raw completion registries for each run were checked against their prescribed ranges and congruence classes. The order-12 run contains all 4,000 indices exactly once, and the order-13 run all 200,000 indices exactly once. For order 14, the 2,000,692 entries cover all 2,000,000 indices, with 692 occurring twice and none more often. The distinct-index file and duplicated-index list in `yuca/logs/` agree with this recomputation. Per-directory summaries contain all 200 directories at order 13 and all 2,000 at order 14, with 1,000 result files each. Summing their counters once per slice reproduces Table 1. After the edge-neighbourhood filter, 218,516 graphs at order 13 and 13,536,597 at order 14 reach the full homology test; the respective outputs contain one and 51 graphs.

Accounting. The valid order-12 array is 2079051 (wrapper 2079049), and the order-13 array is 2079059 (wrapper 2079058). Their task spans are 7 min 10 s and 7 h 10 min 11 s. The valid order-14 arrays, 2079102 on p029 and 2079156 on p011, span 11 October 2026, 11:23:46, to 12 October, 22:26:34, in Hermosillo time (UTC−7): 35 h 2 min 48 s. Up to 896 cores were allocated, with the second front starting later. Allocated core time must not be interpreted as measured CPU time or as a separate benchmark for the generator and filter.

The cardinality rows were computed by separate runs in table mode. The second order-12 run (array 2084878, 4,000 slices of `geng -c 12`, 448 cores, 8 min) computed all columns and reproduced the six columns of the first run exactly. The order-13 row used arrays 2084597 (p029) and 2084636 (p011), with 200,000 slices of `geng -c 13` on up to 896 cores, from 12 October, 21:04, to 13 October, 17:59 (UTC−7). Its registries cover all slices, 483 of them twice. The connected counters, summed once per slice, equal the number of connected graphs on 13 vertices (OEIS A001349). This agreement is a consistency check. It would show that no slice is missing or truncated only under premises that the archived run does not certify: that the residue classes partition the connected graphs, that the stream of each slice contains each graph of its class at most once and no other graph, that the counters are intact, and that exactly one result is summed per residue. Under these premises, writing $N_r$ for the size of class $r$ and $n_r$ for the count observed, $0 \le n_r \le N_r$ and $\sum_r n_r = \sum_r N_r$ force $n_r = N_r$ for every $r$; without them, omissions and duplications could compensate each other in the total. The recomputation described below checks these premises slice by slice.

Correction to the archived worker. In commit `4a059b5` the file named `yuca/as_run/tarea_n14_p011.sh` has SHA-256

`50477550293c8442ee6368b360dbeade20144fe8f20be54f32571d263018b320`

and belongs to the earlier failed attempt with `geng` 2.7r4 (array 2079068), not the successful descending p011 front. Since commit `892e41b` that file is kept unchanged as `yuca/as_run/tarea_n14_p029_intento_fallido_2079068.sh`, and the name `tarea_n14_p011.sh` holds the worker identified for the successful front, found in the primary provenance archive; its 927 bytes have SHA-256

`b95a125b466b47e921269baa43028983ea00081563ae3d4a126fc86452381ba9`.

A copy accompanies the manuscript sources as `appendix_data/tarea_n14_p011_correct.sh`. SLURM accounting on YUCA does not retain submitted batch scripts, so the identification is a forensic corroboration from file-system records, not the recovery of an authenticated copy of the submitted script: this file was created at the second at which array 2079156 was submitted (11

**Algorithm 4:** Distributed search on the cluster

**Input:** $n$; number of slices $M$; number of array tasks $T$; `geng` options $\Omega$; mode (stuck / table)

A one-core wrapper job compiles the program, creates the registries `listas_t`, `enc_t` ($t = 0, \ldots, T-1$) and submits a SLURM array of $T$ tasks with 64 cores each;

**foreach** *task* $t$ *in parallel* **do**

  **foreach** *slice* $r \equiv t \pmod{T}$, $0 \le r < M$, *with* 64 *slices in flight* (`xargs -P 64`) **do**

    **if** $r \in$ `listas_t` **then** skip (resumable);

    run `geng` $\Omega$ $n$ `-q` $r/M$ `|` `atascadas --modo` mode, writing the output to a per-slice file;

    under the lock used by that worker, append the output to `enc_t` and $r$ to `listas_t`;

The wrapper polls progress every 60 s; when all tasks end it sums the counters, verifies that $\bigcup_t$ `listas_t` $= \{0, \ldots, M-1\}$, lists duplicated slices, and records `sacct` accounting;

October 2026, 15:40:09, UTC$-7$) and has not been modified since (equal birth, modification and change times); the task logs of that array print the descending starting residues that this script computes; and no file in the directory of the valid run predates that run, while the failed attempt is preserved in a separate directory. The failed run cannot be certified by its SLURM completion status alone: it lacked pipeline error propagation for the unsupported generator option.

Duplicated slices and remaining certification work. The two valid fronts shared completion registries and per-slice result paths, but used different lock files. Consequently the recorded coverage is complete while the archive does not preserve two independently comparable outputs for each duplicate. The assertion that all duplicated outputs were identical is therefore not retained. Two recomputations close these gaps: a complete rerun of the order-14 search with a pipeline that checks the exit status and the graph count of both processes of every slice and publishes one receipt per slice atomically, and complete-state recomputations of the order-12 and order-13 rows. Until they are completed, the completeness claims remain conditional on the stated hypotheses and are not presented as fully certified computations. No error in the displayed finite certificates has been inferred from these implementation limitations.

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